



Exploiting the Index Effect on OSEBX using Machine Learning

*Trading on predictions made by GLM and XGBoost with a
conditional posterior probability threshold*

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Abstract

In March and September of each year, Euronext decide which companies should be included on the Oslo Stock Exchange Benchmark Index (OSEBX). OSEBX consists of 50-80 companies, all selected from the approximately 200 companies on the Oslo Stock Exchange All Shares Index (OSEAX). We find that new additions and deletions to OSEBX experience significant price effects leading up to the actual date these changes take place, the effective date (ED). These price effects are named the *index effect*.

We use the machine learning (ML) models *eXtreme Gradient Boosting* (XGBoost) and *Generalised Linear Model* (GLM) to predict index composition to OSEBX in the months leading up to ED. We find that both XGBoost and GLM can predict index composition with accuracy higher than 94%, 30, 60, and 100 days in advance of ED.

Next, we simulate portfolios from 2010 to 2022, buying predicted additions and selling predicted deletions. We find that GLM models predict few, but high-yielding companies. XGBoost models predict more additions and deletions and create more diverse portfolios. The best GLM and XGBoost portfolios outperformed OSEBX by respectively 0.95% and 0.32% per month (11.4% and 3.84% per year) in the period from 2010 to 2022. Even after adjusting for risk in a Fama-French 3 Factor Model (FF3), the same portfolios showed significant alphas at a 95% confidence level.

Lastly, we investigate if the same active trading strategy can yield excess returns in an enhanced index portfolio. In practice, we did this by combining the already simulated portfolios with OSEBX, where we optimised the active share of the portfolio to give the combined portfolio a tracking error of 2%. For the enhanced index portfolios, the best GLM and XGBoost portfolios outperformed OSEBX by respectively 0.05% and 0.06% per month (0.6% and 0.72% per year). However, after adjusting for risk factors in FF3 only one of the XGBoost portfolios showed a significant alpha ($p\text{-value} < 0.1$).

In short, we find that ML models can predict upcoming changes to OSEBX with high accuracy and that exploiting the index effect using ML can yield excess returns.

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1 Introduction

The last 20 years have seen a massive increase in flows from *active investing* strategies trying to outperform a given benchmark index, towards *passive index investing* strategies trying to replicate a benchmark index (JP Morgan, 2023). In this era, new index investing methods have emerged (Jorison, 2002). *Enhanced index investing* is one such method, seeking to replicate the benchmark risk while outperforming the benchmark return (Investopedia, 2023). In this pursuit, enhanced index funds have found positive excess returns from exploiting the price effects of stocks being added or deleted from their benchmark index (NBIM, 2020). These price effects were first observed by Shleifer (1986) on the benchmark index S&P 500, and subsequently named the S&P phenomenon, or the *index effect*. Since the discovery of the index effect by Shleifer (1986), it has been investigated further (Lynch & Mendenhall, 1997), and observed on other benchmark indices (Afego, 2017). In Figure 1.1 we illustrate the average price movement of additions and deletions from 2002 to 2023 on the Oslo Stock Exchange Benchmark Index (OSEBX).

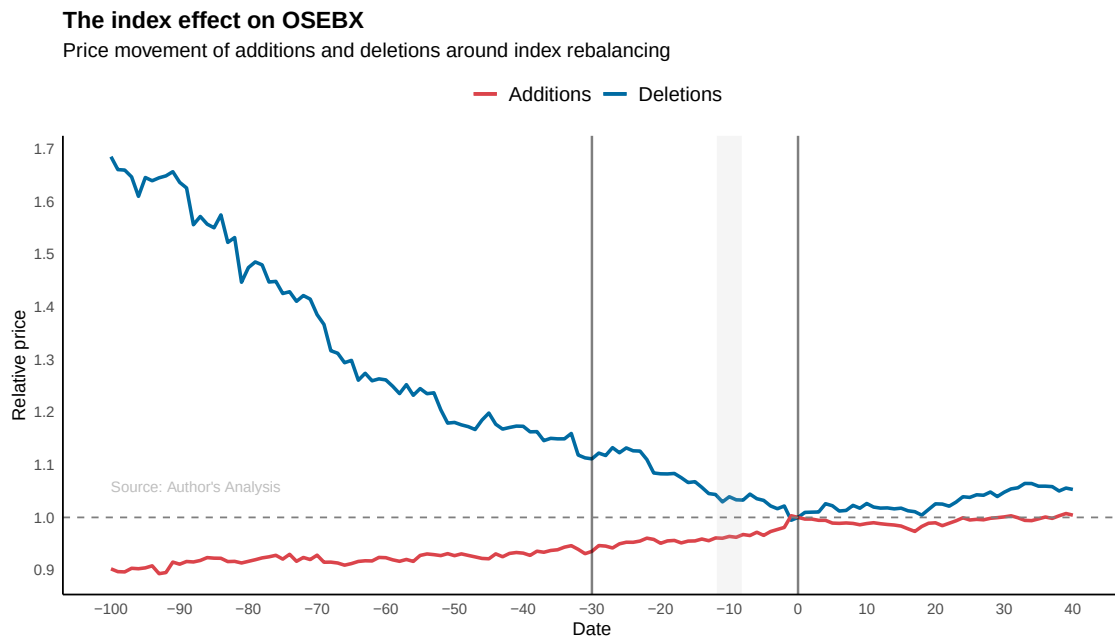


Figure 1.1: The Index Effect on OSEBX

Illustration of aggregated relative price movements to the first effective day (day 0) with additions and deletions for OSEBX from 2002 to 2023. Day 0 represents the first day after the effective date (ED). The announcement date is typically 14 days ahead of the ED, and the data cutoff date (the day data is extracted for each company, which determines the index composition) is about 30 days in advance of the ED. The figure is generated using price data from Bloomberg.

Similar to Shleifer (1986), Figure 1.1 shows that the stock price of additions typically increases in the period before being added to the benchmark index, with the equivalent opposite effect for deletions. There have also been some master's theses indicating the existence of such an effect on OSEBX, among those by Mæhle and Sandberg (2015) and Melingen and Brennmoen (2018). In short, Figure 1.1 shows our illustration of the index effect on OSEBX since 2002.

In March and September of each year, Euronext decides which companies should be on OSEBX. This event is guided by three dates, the cut-off date (CD), announcement date (AD), and effective date (ED) (Oslo Stock Exchange Euronext, 2021). The CD represents the day Euronext extracts data to use for the index rebalancing event. The index composition is announced on AD and implemented on ED. There are typically a few weeks between CD-AD and AD-ED. Figure 1.2 illustrate returns on OSEBX additions and deletions in the 50 days surrounding ED, indicating abnormal returns on ED.

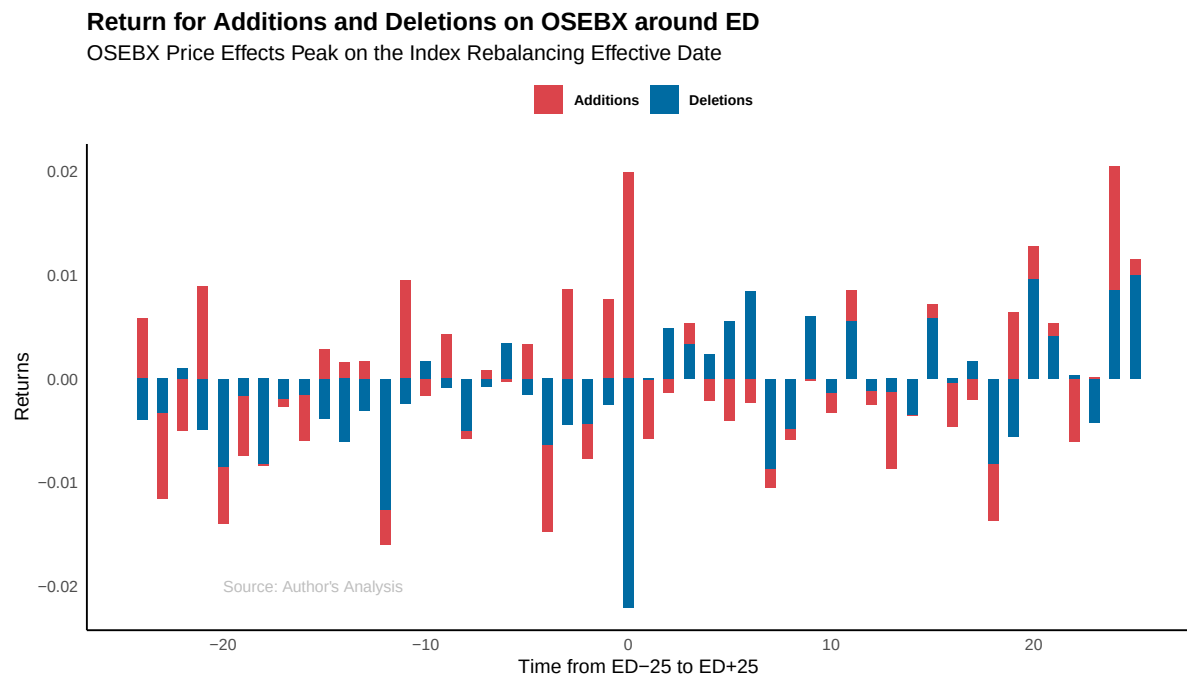


Figure 1.2: Return for Additions and Deletions on OSEBX around ED

We illustrate the aggregated returns for additions and deletions to OSEBX from 2002 to 2023. The figure indicates abnormal returns for both additions and deletions on ED. The return on additions/deletions seems to rise/fall before ED, and somewhat readjust after ED.

In this thesis, we use machine learning (ML) to predict upcoming *additions* and *deletions* to OSEBX ahead of AD. Additions are companies not currently on OSEBX, which enter

from one period to the next, and deletions are companies exiting OSEBX from one period to the next. By predicting before the AD, we can potentially obtain a competitive edge against competing investors who rely on the information announced on the AD. Then, we simulate a trading period from 2010 to 2022, which includes 22 distinct index rebalancing events. Figure 1.3 illustrate the changes we seek to capture. We use *eXtreme Gradient Boosting* (*XGBoost*) (Chen & Guestrin, 2016) and a *Generalised Linear Model* (*GLM*) (Nelder & Wedderburn, 1972) to predict index composition in the next period. Next, we proceed with some definitions and brief reviews of the most common terms used in this thesis.

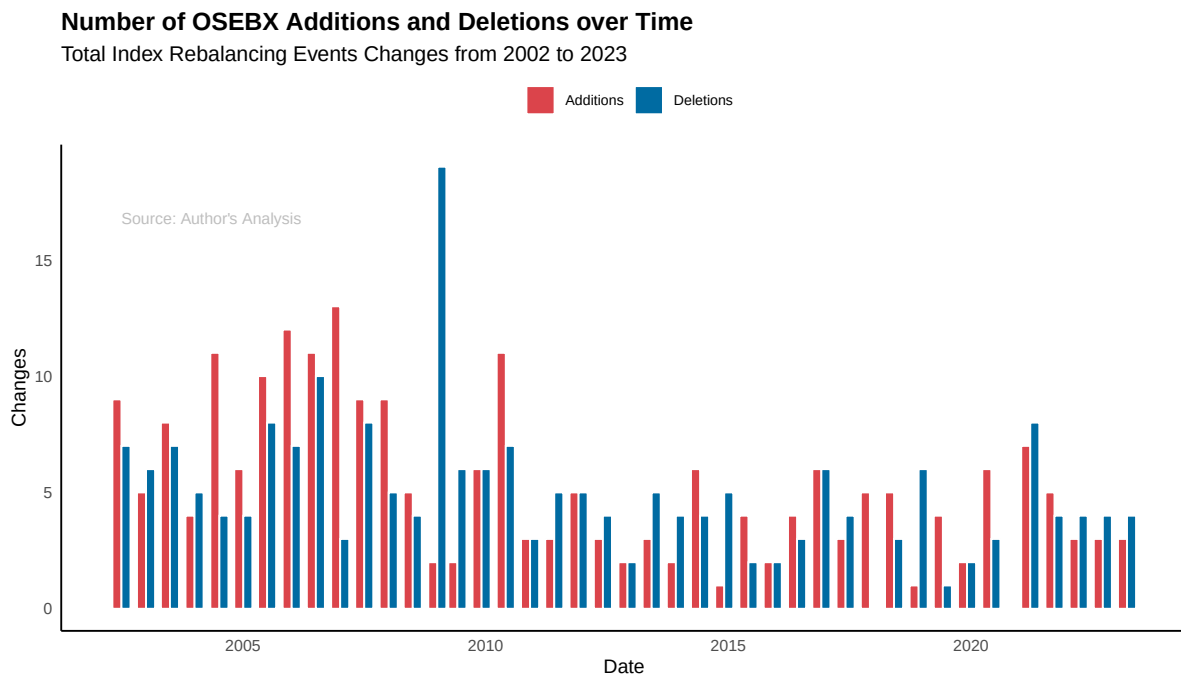


Figure 1.3: Number of OSEBX Additions and Deletions Over Time

Illustration of the number of additions and deletions to OSEBX in the period from 2002 to 2023. Typically, there are between 5-10 additions and deletions per index rebalancing event. We see quite a large number of deletions from OSEBX during the financial crisis.

The world saw its first index investing fund with John Bogle and the Vanguard 500 Index Fund in 1975 (Vanguard, 2023). It was first of its kind in replicating a benchmark index and was mocked as Bogle's Folly. Since then, the fund has seen redemption by generating exceptional returns (Vanguard, 2023). A benchmark index is a grouping of carefully selected assets, to represent the movement of the entire stock market (Goel et al., 2018).

Afego (2017) find differences in benchmark index transparency. The low transparency

S&P 500 has an undisclosed replacement pool and performs need-based rebalancing (Afego, 2017). OSEBX, on the other hand, is rebalanced twice a year and it uses the replacement pool Oslo Stock Exchange All Share Index (OSEAX). The OSEBX composition is determined by size and liquidity criteria, which is publicly available on the Euronext website (Oslo Stock Exchange Euronext, 2021). OSEBX is therefore considered to be relatively transparent (Afego, 2017).

Financial Times (2022) writes that money is pouring out of active funds and into passive, with passive assets under management (AUM) surpassing active AUM in the US. Morningstar (2023) find negative inflow to active funds for 7 of the last 10 years, with over \$250B inflow to passive funds in the same period. JP Morgan (2023) also found \$3T in flow from active to passive equity funds globally in the last 10 years.

Passive index investing has increased in popularity, but it has also received criticism. Ganti and Lazzara (2018) state that "dangers of passive investing" turned up several times more than "dangers of passive smoking" on Google News results in 2018. Criticism of passive investing includes contributions to market bubbles and reduction in market efficiency (Ganti & Lazzara, 2018). This rise of passive investing increases attention toward making enhancements by exploiting the index effect (Elton et al., 2022).

The index effect may emerge from passive funds replicating benchmark indices (Afego, 2017). This can lead to price pressures around ED when the benchmark index changes its composition (Investopedia, 2022). If so, companies are not bought on good underlying information (Kasch & Sarkar, 2012). Peterson (2021) find a revamping in the index effect amidst the trend towards passive investing.

NBIM (2020) has conducted enhancements to their index investing by exploiting the index effect. Using the benchmark index methodology, NBIM predicts and trades on additions and deletions *at* the AD (NBIM, 2020). This strategy significantly boosted the returns of Norway's Government Pension Fund Global from 2000 to 2011 (Ang et al., 2014). Following enhancements, the fund may align more closely with an enhanced index fund than a purely passive one (Bauer et al., 2022).

Defining *enhanced index investing* can be difficult (Riepe & Werner, 1998). In relative

terms, lower tracking error (TE) and higher returns are common characteristics (Riepe & Werner, 1998). Enhancements to index investing, such as exploiting the index effect may justify the definition (Goel et al., 2018). Elton et al. (2022) have found that making such enhancements can outperform passive index investing both for management ability (before costs), and investor returns (after costs). The last years have seen the emergence of global enhanced index funds (JPMorgan Asset Management, 2023). Norway is starting to see this trend, with DNB Asset Management having started an enhanced index fund this October (Furuseth, 2023).

The existence of an index effect on OSEBX has been discussed in previous master's theses, but there is a lack of research on exploiting the index effect in practice. Furthermore, we have seen large investors make enhancements to index investing on large global indices, such as the S&P 500, but not to the same extent on OSEBX. In this thesis, we seek to exploit the index effect on OSEBX in the most practical applied sense possible. We use ML to predict upcoming additions and deletions *before* AD and simulate portfolios trading on these predictions. First, we simulate an active trading portfolio, and thereafter an enhanced index portfolio. We ask the following research questions:

1. *To what degree can machine learning algorithms predict the index composition of OSEBX in the next period?*
2. *To what degree can trading on predicted additions and deletions outperform OSEBX in an active trading portfolio?*
3. *To what degree can trading on predicted additions and deletions outperform OSEBX in an enhanced index portfolio?*

Next, the thesis includes a section on financial portfolio theory in [section 2](#), ML methodology used to predict index composition in [section 3](#), and an explanation of our data set for OSEAX companies in the period from 2006 to 2023 in [section 4](#). In [section 5](#) we assess model performance, and present our findings from the portfolio simulations. Finally, we conclude the thesis in [section 6](#).

2 Theoretical Background

2.1 Introduction to Theoretical background

In this section, we introduce the efficient market hypothesis (EMH) and discuss the index effect in this context. Furthermore, we present the foundation for financial metrics used in evaluating our portfolio simulations in [section 5](#). We discuss the financial metrics alpha α and beta β both in the context of the Capital Asset Pricing Model (CAPM) and the Fama-French 3 Factor Model (FF3). We also look at more general financial metrics, such as return R_p , risk σ_p , Sharpe ratio (SR), tracking error (TE), and information ratio (IR).

2.2 Efficient Market Hypothesis

Samuelson ([1965](#)) posited the key principle of EMH in 1965, if a price movement could be predicted with certainty, it would have already occurred. The principle was furthered by E. Fama ([1970](#)), who argued that security prices in EMH always fully incorporate available information. He categorised EMH into three forms, weak (reflecting historical prices), semi-strong (including all public information), and strong (encompassing all information, including insider insights) (E. Fama, [1970](#)).

2.3 Index Effect

According to EMH, stock prices reflect values of the underlying company (E. F. Fama, [1991](#)). Jain ([1987](#)) point out that announcements on AD of additions and deletions to S&P 500 impact the publicly perceived investment appeal of stocks. If so, there exist price effects that do not reflect the underlying company (Jain, [1987](#)). The index effect challenges EMH, which can have implications for index investing funds.

Index funds must periodically adjust their portfolio to replicate their benchmark index (Investopedia, [2023](#)). Research by Blitz et al. ([2010](#)) and Madhavan and Ming ([2002](#)) find that the timing of these adjustments can significantly impact fund performance and costs. Blitz et al. ([2010](#)) find that adjusting portfolios during certain months outperform others. Madhavan and Ming ([2002](#)) find that spreading trades between AD and ED can reduce costs without major increased risk. If the timing of index rebalancing influences stock

prices, this can challenge EMH (E. Fama et al., 1969).

Afego (2017) find some possible reasons for the existence of the index effect, divided into demand and information-based hypotheses. Demand-based hypotheses view the effect as either temporary, where prices revert after the index rebalancing event (price pressure hypothesis), or permanent (imperfect substitute hypothesis) (Afego, 2017). Information-based hypotheses suggest that index rebalancing signal information to investors, additions/deletions imply positive/negative news (information hypothesis), increased liquidity affect prices (liquidity hypothesis), awareness of index changes affects behaviour (awareness hypothesis), and changes represent rule book criteria (selection criteria hypothesis) (Afego, 2017). There are several hypotheses for why the index effect exists. In this thesis, we are mainly interested in the fact that it *does* seem to exist on OSEBX, as shown in Figure 1.1.

2.4 Financial Metrics

2.4.1 Risk and Return

Markowitz (1952) laid the fundamental groundwork for understanding the trade-off between minimising risk σ_p and maximising return R_p in investment portfolios. The risk σ_p of a portfolio with return R_p is shown in Equation 2.1.

$$\sigma_p = \sqrt{Var[R_p]} \quad (2.1)$$

2.4.2 Capital Asset Pricing Model

Building on portfolio theory work by Markowitz (1952), CAPM was developed independently by economists such as Sharpe (1964) and Mossin (1966). CAPM calculates an asset's expected theoretical return $E(R_i)$. It combines the risk-free rate R_f , the asset's beta β_i indicating its relative risk, and market return minus risk-free rate $E(R_m) - R_f$ to represent the market premium. Equation 2.2 expresses $E(R_i)$ in CAPM, and Equation 2.3 the abnormal return AR_i .

$$E(R_i) = R_f + \beta_i \cdot [E(R_m) - R_f] \quad (2.2)$$

$$AR_i = R_i - E(R_i) \quad (2.3)$$

2.4.3 Fama-French 3 Factor Model

In an extension of CAPM, E. F. Fama and French (1992) introduced new size and value risk factors, for a more accurate assessment of asset pricing and abnormal returns. FF3 recognises over-performance by small-cap (SMB) and high book-to-market (HML) ratio assets (E. F. Fama & French, 1992). FF3 considers the market return R_m , risk-free rate R_f , portfolio alpha α , and sensitivity to market $\beta_{i,M}$, size $\beta_{i,SMB}$, and value $\beta_{i,HML}$. The theoretical expected return $E(R_i)$ is shown in Equation 2.4.

$$\begin{aligned} E(R_i - R_f) &= \alpha_i + \beta_{i,M}(R_M - R_f) \\ &\quad + \beta_{i,SMB} \cdot SMB \\ &\quad + \beta_{i,HML} \cdot HML + \epsilon_i \end{aligned} \quad (2.4)$$

In both CAPM and FF3, the cumulative return CR from time τ_1 over τ_2 periods can be seen in Equation 2.5.

$$CR(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} R_t \quad (2.5)$$

2.4.4 Alpha

Alpha (α) is a key metric in finance, quantifying how an investment performs relative to the market (Sharpe, 1964). It is the excess return of an investment over the predicted return based on its risk β . Positive alpha means the investment outperforms the market after adjusting for risk, while negative alpha indicates under-performance. It's calculated in CAPM and FF3 respectively in Equation 2.6 and Equation 2.7.

$$\alpha_{CAPM} = R_i - [R_f + \beta_i(R_m - R_f)] \quad (2.6)$$

$$\begin{aligned}
\alpha_{FF3} = R_i - [R_f + \beta_{i,M}(R_m - R_f) \\
+ \beta_{i,SMB} \cdot SMB \\
+ \beta_{i,HML} \cdot HML]
\end{aligned} \tag{2.7}$$

2.4.5 Beta

Beta (β) is a crucial financial metric that measures the volatility, or systematic risk, of an investment in relation to the overall market (Sharpe, 1964). It indicates how much an investment's price is expected to fluctuate compared to market movements. Generally, a beta greater than 1 implies higher volatility than the overall market, while a beta less than 1 indicates lower volatility. However, if a portfolio has a negative correlation with the market in some periods, this can reduce the beta even though the portfolio is more volatile than the market. It is shown in CAPM in Equation 2.8. In FF3 one also uses the factors, as shown in Equation 2.9 for SMB. $Cov(R_i, R_m)$ is the covariance of the investment's return with the market return, and $Var(R_m)$ is the variance of the market return.

$$\beta_i = \frac{Cov(R_i, R_m)}{Var(R_m)} \tag{2.8}$$

$$\beta_{i,SMB} = \frac{Cov(R_i, SMB)}{Var(SMB)} \tag{2.9}$$

2.4.6 Sharpe Ratio

SR is a measure of risk-adjusted return. In other words, it describes how much return an investment generates per unit of risk (Sharpe, 1964). Equation 2.10 show the SR, with the return of the investment R_i , with the risk-free rate R_f , and standard deviation σ_i .

$$SR = \frac{R_i - R_f}{\sigma_i} \tag{2.10}$$

2.4.7 Tracking Error

TE is a measure used to indicate how closely a portfolio follows a benchmark index (Roll, 1992). Equation 2.11 shows the TE calculated as the standard deviation of the difference

between returns of the portfolio R_p and the benchmark index R_b . Equation 2.12 show going from monthly to annual TE. DNB Global Enhanced Index (2023) and NBIM (2020) have a TE limit of 0.25-1.00% and 1.00-1.50% respectively.

$$\text{TE} = \sqrt{\text{Var}[R_p - R_b]} \quad (2.11)$$

$$\text{TE}_{ann} = \text{TE}_{mo} \cdot \sqrt{12} \quad (2.12)$$

2.4.8 Information Ratio

Moreover, some active and passive funds find IR to be a more useful metric than TE (Jorison, 2002). IR measures a portfolio's excess return R_p compared to the returns of its benchmark R_b , relative to TE, as shown in Equation 2.13 (Goodwin, 1998).

$$\text{IR} = \frac{R_p - R_b}{\sqrt{\text{Var}[R_p - R_b]}} \quad (2.13)$$

2.5 Summary of Theoretical Background

In Table 2.1 we summarise the financial metrics used to assess our portfolio simulation performance in section 5. Looking at a single financial metric is not optimal, as there is often a trade-off between them. For example, the most fundamental trade-off in finance is between risk and return. We will discuss this further in section 5, where we find certain portfolios outperforming in single metrics, but not in others. In an active trading portfolio, return R_p or SR might be the most important metric.

Metric	Range
R_p	Low (worse) – High (better)
σ_p	Low (better) – High (worse)
α	Worse < 0 < Better
β	Defensive < 1 < Aggressive
SR	Worse < 1 < Better
TE_{ann}	Passive index investing $\approx$ 0-1% Enhanced index investing $\approx$ 1-2% Active investing $\approx$ 2-10%
IR	Worse < 0 < Better

Table 2.1: Summary of Financial Performance Metrics

In principle we are looking for as high as possible values for R_p , α , SR, and IR, and as low as possible values for σ_p and TE. TE and IR become more important in an enhanced index portfolio, which we will discuss in section 5.

3 Methodology

3.1 Introduction to Methodology

In this section, we first introduce ML, and its learning paradigms such as supervised learning (SL), unsupervised learning (UL), and reinforcement learning (RL). We discuss our research questions in the context of a classification problem using SL. Then, we introduce the models used in this thesis, namely logistic regression using GLM, and XGBoost. Next, we introduce metrics used to assess model performance in [section 5](#). We introduce the confusion matrix and derived metrics, such as accuracy, precision, sensitivity, specificity, and F-score. We also describe descriptive plots as area under operating characteristic curve (AUROC) and variable importance plots (VIP).

3.2 Machine Learning

Samuel ([1967](#)) is accredited with introducing the term ML. He states that ML is a field of study that makes computers learn without explicitly being programmed (Samuel, [1967](#)). ML described the pattern recognition tasks from the learning component of artificial intelligence (AI), which in itself was introduced by McCarthy ([1955](#)). Samuel ([1967](#)) applied ML to the game of checkers, a game having clear rules and motivations. Following breakthroughs in ML (Sutskever et al., [2012](#)), and AI during the last decade (Vaswani et al., [2017](#)), the world has experienced massive increases in the application of ML. This is especially shown through tools such as large language models (OpenAI, [2023](#)).

Hastie et al. ([2001](#)) describe ML as a set of methods for identifying patterns and trends in large data sets. This is expanded by James et al. ([2023](#)), who discusses statistical learning as a tool for interpreting complex data. ML is mainly divided into SL, UL, and RL (James et al., [2023](#)). SL focuses on predicting an output variable from input variables, while UL aims to uncover patterns and associations among input variables without a specific output variable (Hastie et al., [2001](#)). In RL, one trains an ML model to move towards something, refining its behaviour through cost functions (Sutton & Barto, [2018](#)). The three types of ML all have useful applications in different contexts (Hastie et al., [2001](#)). In this thesis, we apply SL when predicting index composition on OSEBX.

White (1988) made an early application of ML in finance, forecasting the daily stock returns of the US company IBM. The following decades experienced massive increases in the application of ML in finance (Warin, 2021). Goodell et al. (2021) divides current themes of ML in finance into (1) portfolio construction, (2) financial fraud, and (3) forecasting. In our context, trading on predicted upcoming additions and deletions using ML fits the first theme (Goodell et al., 2021).

3.3 Logistic Regression

Regression analysis is a statistical method for predicting a dependent variable based on one or more independent variables, often used to determine relationships and trends (Galton, 1886). Cox (1958) furthered the concept to binary sequences, classifying outcomes as success/failure, or 1/0. This concept forms the basis of classification problems (Cox, 1958). Using ML to assign new input vectors of explained variables to a finite number of target categories is named a classification problem (Bishop, 2006). If the target output is comprised of continuous variables, this is a regression problem (Bishop, 2006).

Regression presupposes ordered relationships between outcomes, such that ordinary least squares (OLS) can be used to fit a model (Hastie et al., 2001). This approach does not make sense when predicting classes, which do not have such linearly ordered relationships (Hastie et al., 2001). Linear models are suitable for regression (Pearson, 1901), but struggle with the binary nature of a classification problem. Logistic regression uses a logistic function to ensure outputs between 0 and 1, representing probabilities (Berkson, 1944). This makes it ideal for classification tasks where outcomes are distinctly categorical (Berkson, 1944).

In logistic regression, a linear combination of independent variables represented as z with coefficients β_0 and β_1 in Equation 3.1, is transformed by the logistic function into a probability. This maps the linear combination to a value in the interval $[0,1]$, as in Equation 3.2. Combining these steps, the logistic regression model is fully expressed in Equation 3.3.

$$z = \beta_0 + \beta_1 X \tag{3.1}$$

$$p(X) = \frac{1}{1 + e^{-z}} \quad (3.2)$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \quad (3.3)$$

Nelder and Wedderburn (1972) introduced GLM, which enhances traditional linear models by using iterative weighted regression for efficient parameter estimation via maximum likelihood. We apply logistic regression using GLM in R (as our main analytical tool), with the function `glm::glm()` (Dobson, 2023). In R, `y` is the target variable, `x` the predictor variables, `family=binomial` the binomial distribution, and `link="logit"` specifying the use of logistic regression.

3.4 Tree Boosting Models

Schapire (1990) first introduced the procedure for boosting models. Boosting models build on an initial model, and use base models to *boost* accuracy over sequential iterations. He showed that weak learners could improve their model performance when training additional classifiers (Schapire, 1990). In this context, a weak learner is an algorithm for producing a two-class classifier that outperforms random chance. The weak learner algorithm improves on an initial classifier h_1 by *boosting* model performance over several iterations. Schapire (1990) was first able to prove that the boosted classifier h_B outperform the initial h_1 in model performance. In Algorithm 1, J. Friedman et al. (2000) illustrate the boosting algorithm by Schapire (1990).

Algorithm 1 Schapire Weak Learner Algorithm (J. Friedman et al., 2000)

Input: Data training set D , number of data points N .

- 1: h_1 is learned on first N training points in D ;
- 2: h_2 is learned on a new sample of N point in D , where half are misclassified by h_1 ;
- 3: h_3 is learned on N points in D , where h_1 and h_2 disagree
- 4: h_B is the boosted classifier, where $h_B = \text{Majority Vote}(h_1, h_2, h_3)$

Output: Boosted classifier h_B

Tree boosting has proven to be a highly efficient ML model in doing predictions (Chen & Guestrin, 2016). Tree boosting builds on intuition by Schapire (1990). Both the base learner and base models are simple. In other words, they typically have high bias and low variance (Nielsen, 2016). Tree boosting start with an initial model and find the final

model by iterating M times over base models. For each iteration m where $m = 1, \dots, M$, tree boosting models seek to minimise a loss function compared to the true target variable in the training data set. Each base model is weighted, which shows their contribution to the final model (Nielsen, 2016).

In this context, a base model can be a simple decision tree model. Tree boosting is somewhat backward compared to classical ML algorithms, as the base models are evaluated compared to the true value of the target variable in training. As such, one could expect tree boosting to produce biased models. There is commonly a trade-off between bias and variance in a model (an Jacob Feldman, 2011), which is illustrated in Figure 3.1. Tree boosting alleviate this by iterating for large values of M , penalising complex models, and scaling down base models by learning rates (or shrinkage).

Bias-Variance Trade-Off

Optimal Model Complexity is in the Intersection between Bias and Variance

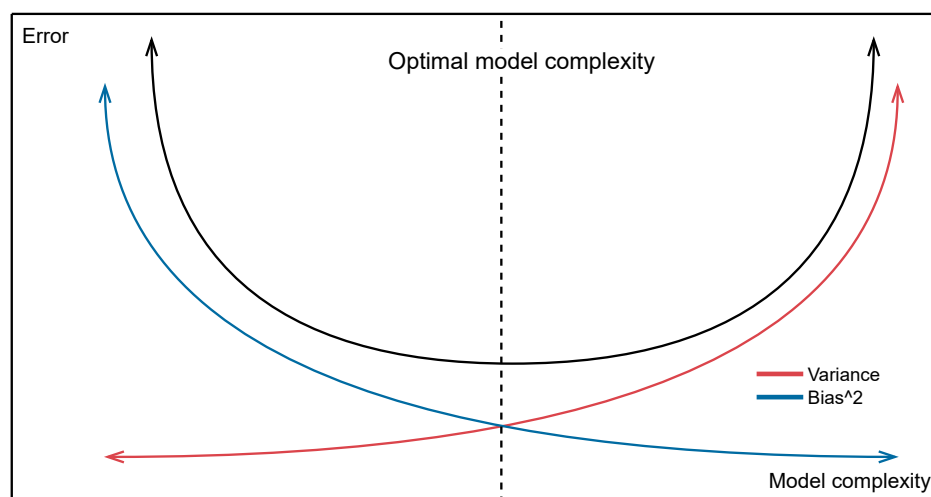


Figure 3.1: Bias-Variance Trade-Off

The model complexity is in a trade-off between bias and variance. Here, over-fitting typically entails high bias and low variance, while under-fitting entails low bias and high variance.

There exist other tree models which can improve the performance of a simple decision tree. Random forest (RF) is one such example, where a "forest" of many decision trees are generated. Here, the prediction made by the final RF model is the average prediction across a single iteration of a single forest. In Figure 3.2 we illustrate a single decision tree model and show the difference between a RF and tree boosting model. Typically, the performance ordering of these are single tree model $<$ RF $<$ tree boosting (James et al.,

2023).

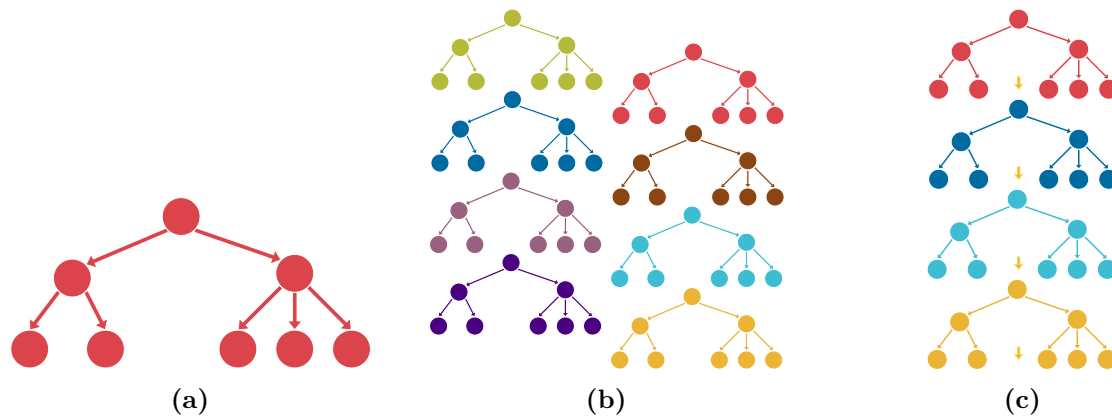


Figure 3.2: From a Single Tree Model (a) to Random Forest (b) to Tree Boosting (c)

We illustrate the simple intuition behind going from a single decision tree toward more complex tree models. RF average across a single iteration of several trees, while tree boosting aggregate over several iterations of a single tree.

3.5 XGBoost

Following tree-boosting algorithms producing competitive and robust classification models, J. H. Friedman (2001) introduced gradient boosting. This is based on gradient descent. In other words, gradient boosting seeks to optimise an algorithm by finding the minimum of a function. It does so by moving opposite to the gradient of the function, or steepest ascent. The move from gradient to hessian boosting includes using not only first-order (gradient), but also second-order (hessian) methods when optimising boosting algorithms (Nielsen, 2016). XGBoost by Chen and Guestrin (2016) is a tree boosting model which utilise both gradient and hessian boosting.

The last years have seen a prominent rise in the popularity of XGBoost, by winning several ML competitions (Nielsen, 2016). XGBoost is renowned for its efficiency and flexibility in handling various data types, and the ability to customise XGBoost has made it an optimal tree-boosting system of choice in ML competitions (Nielsen, 2016).

The function `xgboost::xgboost()` is an application of this in R (Yuan, 2023). The documentation builds on work by J. Friedman et al. (2000) and (2001). XGBoost comes with a wide range of configurable hyperparameters, which can be used to utilise the

full potential of the model (kaggle, 2020). The hyperparameters might be optimised to increase a specific measure, but in our application, we have used a set of parameters without optimising. This is mainly due to the low complexity of our classification problem. Hyperparameters used in this thesis are shown in [Appendix F](#).

3.6 Posterior Probability Threshold

The ML models output an estimated likelihood of which class the observations belong to, but the actual classification is determined based on the likelihood being above or below a certain threshold. This threshold is often set to 0.5, but in some practical applications, the predictions can be improved by using a different threshold (James et al., 2023). As James et al. (2023) mentions, when predicting if a bank customer is going to default or not, one can adjust the threshold to increase accuracy. James et al. (2023) refer to this threshold as the *posterior probability threshold (PPT)*. In short, PPT can be adjusted to accommodate for specific use cases and increase accuracy. In this thesis, PPT can influence the distribution of the predicted additions and deletions to OSEBX, and also the total accuracy. In [section 5](#), we carefully consider which PPT should be applied in this thesis.

3.6.1 Conditional Posterior Probability Threshold

In our classification problem, it becomes evident that it is difficult for the ML algorithms to capture when a company is added or deleted from OSEBX. This is a practical challenge because we are dependent on predicting additions and deletions to exploit the index effect. We have considered several approaches to cope with this difficulty. First, we considered (1) implementing a custom objective function in the XGBoost application programming interface (API), where XGBoost enables for customising a loss function and a corresponding evaluation measure. The idea here would be to create a custom objective function to penalise model behaviour when not predicting changes. A different approach is to (2) implement a *conditional PPT*, based on the status of the company in the previous period. The second approach considered is fairly easy to implement, and we investigate this approach further.

Seeking to increase the number of predicted changes made by the ML model, we add a

lower PPT for companies not currently on the index (potential predicted additions), and a higher PPT for companies currently on the index (potential predicted deletions). To apply this in practice, let $Y_{i,t}$ represent the class for company i at time t . Then $Y_{i,t-1}$ represents the class for company i in the previous period $t - 1$. By knowing the $Y_{i,t-1}$ for a specific observation, we can use this information to decide which PPT we would like to apply for that specific classification. When $Y_{i,t-1}$ is equal to 1, we know that the company was listed on OSEBX in the previous period. To increase the total number of predicted deletions, we would therefore use a higher PPT for this classification. In contrast, if $Y_{i,t-1}$ is equal to 0, we would like to lower the PPT, because then we would predict more additions to OSEBX. Next, we introduce a parameter ε which is either added or subtracted from the PPT. We also define θ as the PPT, and θ^* to be the conditional PPT. The conditional PPT can then be expressed as in [Equation 3.4](#).

$$\theta^* = \begin{cases} \theta + \varepsilon, & Y_{i,t-1} = 1 \\ \theta - \varepsilon, & Y_{i,t-1} = 0 \end{cases} \quad (3.4)$$

By using this equation, we must also define the valid values for θ^* , which can be expressed as $0 < \theta^* < 1$. When classifying the observations, the predicted class $\hat{Y}_{i,t}$ is 1 if the estimated probability p is greater than θ^* , and 0 if p is smaller than θ^* . To interpret [Equation 3.4](#) in words, we can describe it as the following: To keep staying on OSEBX will require a higher estimated probability, than entering OSEBX will require. Vice versa, to keep staying off OSEBX will require a lower estimated probability, than exiting OSEBX will require.

However, by using this approach, we encounter some important considerations. First, if a higher number of trades is better at capturing the index effect, why not buy all companies currently not on OSEBX, and sell all companies currently on OSEBX. By doing so, we would be trading all of the approximately 200 constituents on OSEAX. Most likely, the exploited index effect would then be significantly diminished. Another consideration would be to decide whether or not the conditional adjustment parameter ε should be symmetrical or not in relation to the PPT. More specifically, we need to decide if the same parameter should be applied in both directions (add and subtract the same parameter from the initial PPT). Because we want to analyse the performance of both long and short

positions in our simulated portfolios, we find it more appealing to treat both additions and deletions equally. We proceed using a symmetrical addend in this thesis. In practice, we are required to use a numerical value of ε . In [section 5](#) we carefully consider which value of ε to implement in our models.

3.7 Machine Learning Performance Measures

3.7.1 Confusion Matrix

A confusion matrix is a useful tool in evaluating and visualising a classification model's performance (Caelen, 2017). In a binary task, the confusion matrix is a 2×2 matrix, where TP is the true positive, FP false positive, FN false negative, and TN true negative, as shown in [Figure 3.4](#) (Caelen, 2017).

$$\begin{bmatrix} TN & FN \\ FP & TP \end{bmatrix}$$

Figure 3.4: Confusion Matrix

Illustration of a confusion matrix, displaying TN (true negatives), FN (false negatives), FP (false positives) and TP (true positives)

3.7.2 Accuracy Metrics

From the confusion matrix in [Figure 3.4](#), we can calculate accuracy, precision, sensitivity, and specificity (Coenen, 2012). Accuracy is calculated as the ratio of correct predictions to the total observations ([Equation 3.5](#)), precision is the ratio of true positives to all predicted positives ([Equation 3.6](#)), sensitivity is the proportion of actual positives correctly identified ([Equation 3.7](#)), and specificity is the proportion of actual negatives correctly identified ([Equation 3.8](#)) (Coenen, 2012). F-score, illustrated in [Equation 3.9](#), provides a balanced measure of model performance in classifying data, and is especially useful in situations when considering both FP and FN (Aydin, 2021).

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (3.5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.6)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (3.7)$$

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (3.8)$$

$$\text{F-score} = \frac{TP}{TP + \frac{(FP + FN)}{2}} \quad (3.9)$$

3.7.3 Area Under Receiver Operating Curve

The receiver operating characteristics (ROC) curve depicts the performance of a classifier in two dimensions, with sensitivity (true positive rate) and 1-specificity (false positive rate) along the axes (Fawcett, 2006). AUROC gives a model performance in the interval from 0 to 1 (Hanley & McNeil, 1982). Figure 3.5 illustrates AUROC, where 0.5 is a random guess, and 1.0 is a perfect classifier.

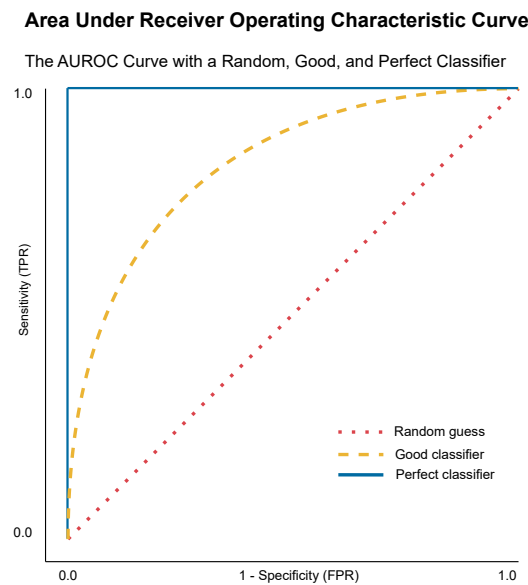


Figure 3.5: The AUROC Curve

The area under receiver operating characteristic curve is shown with a random guess, a good classifier, and a perfect classifier.

3.7.4 Variable Importance Plot

ML models have commonly been evaluated with only a single metric, such as accuracy (Koalaverse, 2023). However, it is increasingly important to not only focus on model predictions but also to understand how these predictions are made (Doshi-Velez & Kim, 2017). VIP shows the relative significance of each predictor variable in an ML model

(Koalaverse, 2023), as shown in Figure 3.6. VIP is calculated differently across ML models, but the core idea is to display the relative importance of predictor variables (Wei, 2015). VIP can be applied in R using the `vip::vip()` function (Greenwell, 2023).

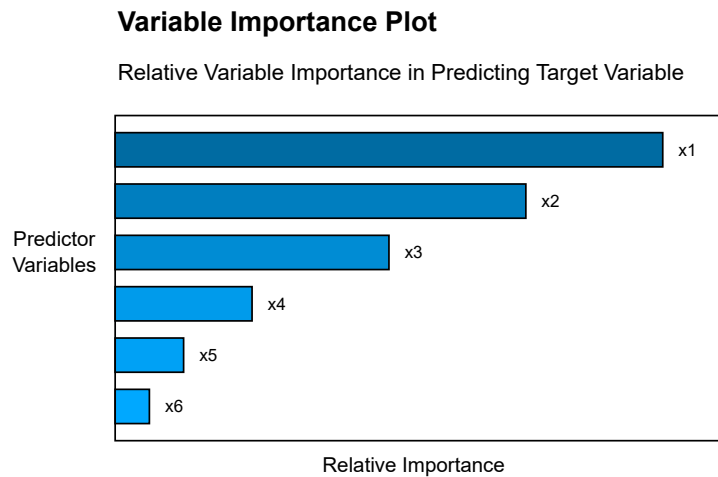


Figure 3.6: VIP

This shows the relative importance of predictor variables in predicting the target variable. Note that these are relative metrics, and absolute numbers will deviate between GLM and XGBoost models, as seen in section 5.

3.8 Summary of Methodology

In this section, we first looked at ML and the application of ML methods in finance. We introduced the ML models GLM and XGBoost, where both will be used in the portfolio simulation in section 5. Further, we also introduced PPT, the conditional PPT, and the ML performance metrics, as summarised in Table 3.1. For the numeric ranges in Table 3.1, values closer to 1 are desirable.

Metric	Range
Accuracy	[0, 1]
Precision	[0, 1]
Sensitivity	[0, 1]
Specificity	[0, 1]
F-score	[0, 1]
AUROC	[0.5, 1]
VIP	Relative

Table 3.1: Summary of ML Evaluation Metrics

In principle we are looking for higher values for the performance metrics accuracy, precision, sensitivity, specificity, F-score, and AUROC. In section 5 we will discuss how only looking at accuracy measures may be misleading depending on the practical application of an ML model.

4 Data

4.1 Introduction to Data

The Euronext index methodology for OSEBX lays the foundation of data used in our analysis (Oslo Stock Exchange Euronext, 2021). Euronext collects data on OSEAX companies on CD, which are used to calculate upcoming index composition. The compositions are publicly announced on AD, and implemented on ED (2021), as shown in Figure 4.1. For OSEBX, this timeline occurs twice a year, as the benchmark index rebalances semi-annually in March and September (2023). In section 5, we use data from January 2006 to March 2023.

We decided to analyse three different prediction horizons so that for each index rebalancing event, we collect data from 30, 60, and 100 business days before ED. For example, for the index rebalancing event in March 2023, where ED is 2023-03-17, we collect data for all OSEAX companies in February of 2023, December and October of 2022.

Timeline of Index Rebalancing on OSEBX

Each distinct Index Rebalancing Event includes CD, AD, and ED

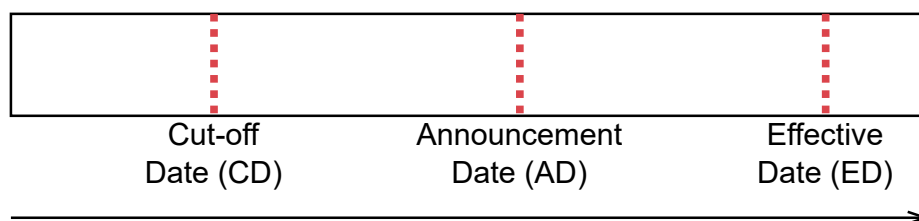


Figure 4.1: OSEBX Index Rebalancing Timeline

Timeline of a single index rebalancing event on OSEBX with CD, AD, and ED. In the portfolio simulation in section 5 from 2010 to 2023 we experienced two distinct index rebalancing events each year (except 2020).

4.2 Variables

Euronext collects four key variables on CD reflecting the liquidity and size of companies (Oslo Stock Exchange, 2023). The first variable explains the proportion of a company's active trading days out of the total trading days in a given period (2021). We call this variable `trading_status`. The second is turnover (electronic order book), which is the total value of shares traded at the end of the last 12 months up to and including CD, but

excluding the 12 days with the highest turnover for each company (2021). We name this variable `turnover_eob`. The next variable is the free float factor, which represents the free float market capitalisation for the total market value of shares that are publicly available for trading (2021). We call this variable `free_float`. The last collected financial variable `icb_supersector` is based on the industry classification benchmark (ICB) according to Dow Jones and FTSE (FTSE Russel, 2023).

In short, Euronext collects the four variables above to calculate index composition, based on the publicly available index methodology (Oslo Stock Exchange Euronext, 2021). We do not review the rule book in greater detail here but in general the bigger `turnover_eob`, `free_float` and `trading_status`, the more likely the company will be included on OSEBX according to Oslo Stock Exchange Euronext (2021). In this thesis, we seek to use the variables Euronext collects and use ML to predict upcoming index composition *before* AD. If we can predict OSEBX changes before they are announced to the public, we may gain a temporal advantage. We seek to make our ML models as simple as possible, and therefore try not to use additional financial variables, which are not mentioned in the OSEBX rule book (Oslo Stock Exchange Euronext, 2021). We consider our findings to be easier to replicate and understand when not using any additional financial variables.

4.2.1 Target Variable

In our data, we introduce the binary target variable `index_cp` to denote if a company is on the benchmark index OSEBX in the current period. This variable is equivalent to $Y_{i,t}$ explained in section 3. The current period represents the time from the last ED to the upcoming ED. As such, we seek to predict `index_cp` for the *next* period. The binary target variable `index_cp` is 1 if the company is on OSEBX, and 0 otherwise. Labeling our data set as such becomes an ML classification problem using SL (Hastie et al., 2001).

4.2.2 Lag Variables

Certain selection criteria differ for companies being on/not on OSEBX (Oslo Stock Exchange Euronext, 2021). Furthermore, we are dependent on knowing the lagged target variable to use the conditional PPT, introduced in section 3, where the lagged variable is denoted as $Y_{i,t-1}$. Therefore, we introduce lag variables showing the binary value for

`index_cp` in the previous periods. We introduce lag variables `index_lag1`, `index_lag2`, `index_lag3`, and `index_lag4` for the value of `index_cp` in the previous 1 to 4 periods. Lag variables can be applied in R with the `stats::lag()` function (RDocumentation, 2023).

4.3 Data Sources

We first collect the dates for relevant index rebalancing events from Oslo Stock Exchange (2023). Thereafter, we collect the benchmark index constituents on OSEAX and OSEBX the day after each ED, to make sure we know the index composition *just after* each rebalancing event. Further, we collect the four financial variables for each company 30, 60, and 100 days in advance of the ED. This is done using a Bloomberg Terminal (BBG) (Bloomberg, 2023b). For the portfolio simulation, we obtain stock prices for all companies using DataStream and a Refinitiv Terminal (DataStream, 2023). Lastly, we obtain FF3 data by Ødegaard (2023a).

We find public announcements on OSEBX constituents by Oslo Stock Exchange (2023) dating back to 2002. In both the BBG and Refinitiv Terminals we are able to find variable and price data for some companies backdating to 2002, but also find significant missing values for other companies in the period before 2006. With this backdrop, we conclude to use data from January 2006 to March 2023, which after data manipulation yields a clean data set with no NA-values. The selected time horizon gives a total of two distinct index rebalancing events each year (except for 2020 when Euronext changed from rebalancing in June/December to March/September), all including a CD, AD, and ED, with the first and last ED being 2006-06-30 and 2023-03-17. All rebalancing dates (CD, AD, and ED for each rebalancing) are shown in Appendix D.

4.3.1 Bloomberg

Specifically, we obtain data from BBG through the BBG Query Language (BQL) (Bloomberg, 2023a), implemented in Excel (Fintools, 2023). We retrieve historical data points through functions such as BDH, BDP, and BQL (Dartmouth, 2023). Please note that the specific BQL functions used are shown in Appendix G. In conclusion, after gathering data from BBG we were left with a data frame, including the following variables DD (the

date the data was collected, 30, 60 or 100 days in advance of ED), `CD`, `AD`, `ED`, `ticker`, `name`, `icb_supersector`, `free_float`, `turnover_eob`, `trading_status` and `index_cp`. Please note that `index_cp` is defined as 1 if the observation's combination of `ticker` and `ED` exists in the data frame where only the OSEBX constituents are stored, which we also obtained from BBG. These variables are collected for each company listed on OSEAX since 2006.

4.3.2 DataStream

Because we want to simulate the historical performance of the constructed portfolios, we need to collect historical price data. In the simulations, we use volume-weighted average price (VWAP). The VWAP is a more accurate representation of the share price we could have bought or sold many stocks for, compared to the closing price (which is often reported when illustrating stock price development). VWAP is also commonly used to evaluate the performance of stock traders (Madhavan, 2002). The VWAP is collected using Refinitiv DataStream of all OSEAX companies in the period from 2006 to 2023 (DataStream, 2023). Collecting these prices is done in practice through the Refinitiv terminal (LSEG, 2023), with an Excel add-in (Refinitiv, 2023). The data frame generated by DataStream consists of one date column, and 487 columns representing each company to ever be listed on OSEAX since 2006. In conclusion, this data source gives us the historical VWAP for each company for each day since 2006.

4.3.3 Fama-French 3-factor Model Data

The FF3 factors on Oslo Stock Exchange (OSE), such as sensitivity to size $\beta_{i,SMB}$ and $\beta_{i,HML}$ is collected from a data file compiled by Ødegaard (2023a). Note that the collected FF3 is only available until 2022-05-31. As such, the actual portfolio simulation in section 5 goes until 2023-03-17, but the portfolio performance assessments use the period from 2010-02-28 to 2022-05-31 due to lack of FF3-factors after May of 2022.

4.3.4 Risk-Free Interest Rate

The risk-free interest rate R_f is the expected return on a zero-risk investment, introduced by Fisher (1930). It is useful when calculating the financial metrics discussed in section 2. The risk-free rates R_f in the Norwegian market are estimated and published by Ødegaard

(2023b). We collect the monthly risk-free rate R_f from Ødegaard (2023b), for the portfolio performance analysis period from 2010 to 2023. Thereafter we average these rates, which amounts to a monthly risk-free rate of $\approx 0.12\%$, or 1.46% annually.

4.4 Data Manipulation and Missing Data

After obtaining the data presented, we still need to make certain operations to make it viable for analysis. We are now referring to the data set obtained from BBG, which ultimately is what will be used for our analysis. When a company is listed on OSEAX after the data cutoff point, we receive NA-values for those companies for the relevant period. This makes it somehow difficult to capture for our ML model if company data is not available at or before CD (or the date of prediction). Vice versa, it is difficult if a company is unlisted from OSEAX between CD and ED. We simply decide to remove rows where data on CD is not available, as this has minimal influence on our data set, and most of the companies entering OSEAX are not eligible for OSEBX the same period, with some exceptions, (Oslo Stock Exchange Euronext, 2021).

Regarding `icb_supersector`, we are missing this variable for some companies before 2010. This stems from companies being classified before the standardisation toward ICB and then delisted. Luckily, we find the Global Industry Classification Standard (GICS) for these companies (MSCI, 2023). Because ICB and GICS overlap, with 25 and 20 level 2 classifications each, we choose to manually estimate ICB for the relevant companies, as shown in [Appendix E](#).

After cleaning the data sets for NA-values, we add the lagged variables `index_lag1`, `index_lag2`, `index_lag3`, and `index_lag4` using the `stats::lag()` function (RDocumentation, 2023) in R.

4.5 Train-Test Split

Splitting the data set into train and test data is an approach used in cross-validating ML models (Geisser, 2012). The idea of cross-validation is that the accuracy of the ML model is evaluated on a data set (test data), that is different than the data set it is trained on (train data). The approach is used to avoid over-fitting, as the test data set is not

leaked to the train data set. To create a train-test split in our case, we use time series cross-validation (TSCV) (Bergmeir & Benitez, 2012). Here we train the models on all historical rebalancing events before the point in time the predictions are made. Originally, we start by using train data up to before 2010, and thereafter add one period of train data for each new index rebalancing event. Figure 4.2 illustrates the split between the train and test data set we use in TSCV. As an example, when making predictions leading up to March of 2023 in Figure 4.2, we train the ML model on all data up to that point.

Time-Series Cross-Validation of Data from 2006 to 2023

The Data is split into Train and Test Set using Time-Series Cross-Validation

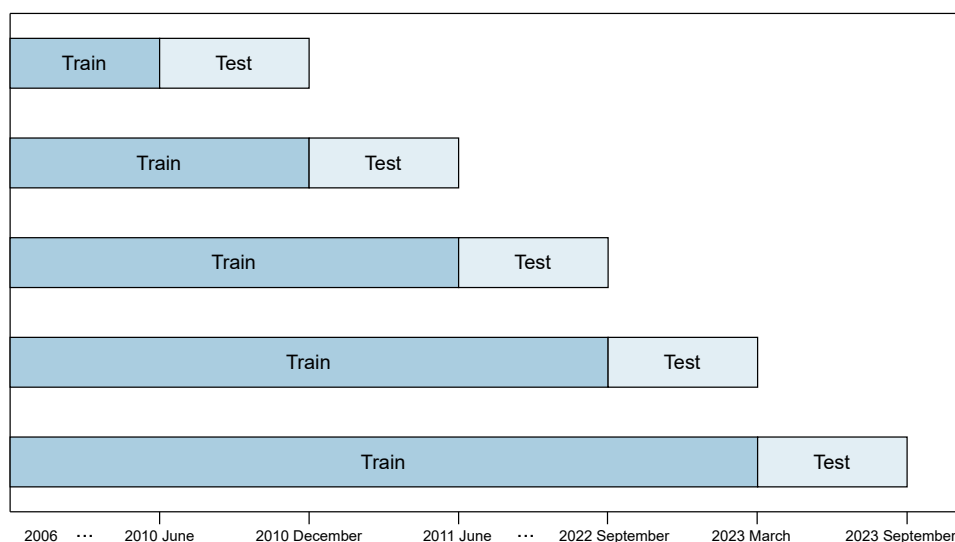


Figure 4.2: Time Series Cross-Validation

TSCV used in portfolio simulation for the trading strategy and enhanced index investing strategy.

4.6 Summary of Data

The final data set used in the ML models is summarised in Table 4.1. In short, we have generated three data sets containing these variables, one data set per prediction horizon of 30, 60, and 100 days before ED. As such, the `trading_status`, `turnover_eob` and `free_float` will be somewhat different between the three data sets. The lagged variables from `index_lag1` to `index_lag4` will be the same across prediction horizons, as the time horizons of 30, 60, and 100 days do not overflow to the previous period. Note that `index_cp` is the target variable, which after prediction indicates if the company will be or not be on OSEBX in the *next* period. The lag variable `index_lag1` indicates if a

company is on OSEBX before the upcoming index rebalancing event. Also note that we use all the predictor variables in [Table 4.1](#) when predicting the target variable `index_cp`. One could probably investigate further if all variables are necessary, and maybe even add other financial variables indicating size and liquidity. In short, we have chosen to keep the number of predictor variables relatively small compared to the amount of data points. The number of rows in the data sets is 5983, 5924, and 5891 for the 30, 60, and 100-day data frames respectively. The difference in the number of rows comes from the fast listings/delistings problem which means we need to discard more rows the longer the prediction horizon.

Variables	Explanation	Abbreviation
Informational variables:		
<code>data_date</code>	Data Date	DD
<code>cutoff_date</code>	Cut-off Date	CD
<code>announcement_date</code>	Announcement Date	AD
<code>effective_date</code>	Effective Date	ED
<code>ticker</code>	BBG Ticker	
Predictor variables:		
<code>trading_status</code>	Share of Days Traded	TS
<code>turnover_eob</code>	Turnover	T0
<code>icb_supersector</code>	ICB	ICB
<code>free_float</code>	Free Float Market Cap	FF
<code>index_lag1</code>	$\text{lag}(\text{index_cp}, 1)$	L1
<code>index_lag2</code>	$\text{lag}(\text{index_cp}, 2)$	L2
<code>index_lag3</code>	$\text{lag}(\text{index_cp}, 3)$	L3
<code>index_lag4</code>	$\text{lag}(\text{index_cp}, 4)$	L4
Target variable:		
<code>index_cp</code>	1 if on OSEBX, 0 otherwise	IC

Table 4.1: The Final Data Set of Predictor Variables and Target Variable

Predictor Variables used in the ML models to predict the target variable `index_cp` for the next period. Informational variables are from OSE Newsweb, financial predictor variables are from Euronext index methodology for OSEBX and lagged predictor variables are calculated.

Before we present our results, we briefly present a descriptive plot to better understand the data set. The number of listed companies on OSEAX and OSEBX is illustrated in [Figure 4.3](#), and we note that there are approximately 50–80 members on OSEBX for a given period, significantly less than those companies not included on OSEBX. There

seem to be only about 5–10 additions and deletions in total for each period, as shown in Figure 1.3.

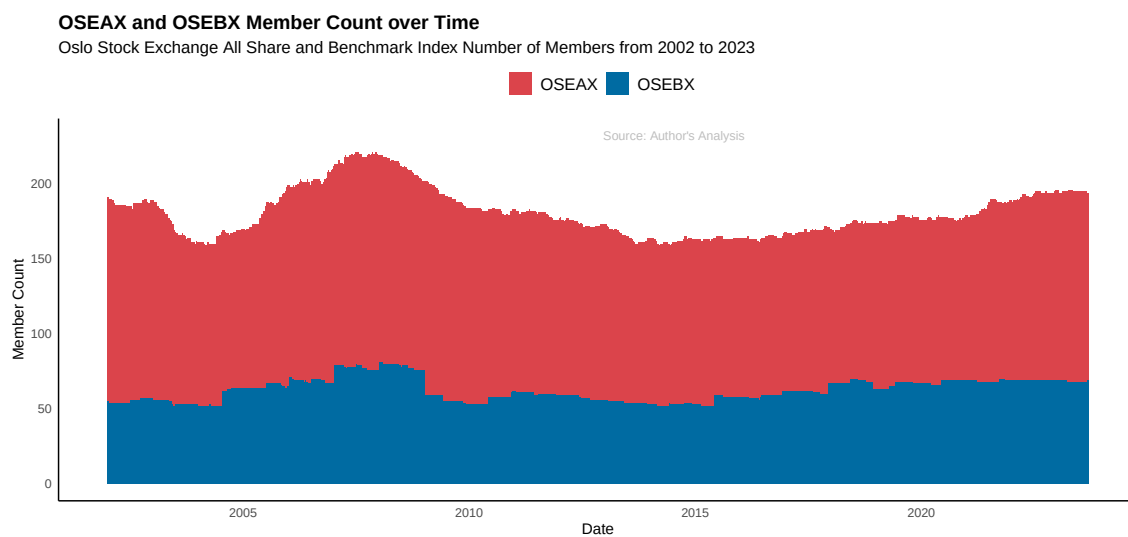


Figure 4.3: Members over Time

All OSEBX companies are also on OSEAX. Notice that OSEAX typically has about 200 companies, whereas 50-80 companies on OSEBX.

5 Results and Discussion

5.1 Introduction to Results and Discussion

In this section we first assess the model accuracy of our predictive ML models, using metrics as described in [section 3](#), and summarised in [Table 3.1](#). This seeks to answer the first research question (1) "To what degree can machine learning algorithms predict the index composition of OSEBX in the next period?". Next, we simulate portfolios in an active trading strategy, where we trade on the predicted additions and deletions made by the ML models. We do so by running a portfolio simulation from 2006 to 2023 and assessing the performance with financial metrics introduced in [section 2](#). This seeks to answer the second research question (2) "To what degree can trading on predicted additions and deletions outperform OSEBX in an active trading portfolio?". Lastly, we repeat the portfolio simulation, but this time in an enhanced index strategy, seeking to answer research question (3) "To what degree can trading on predicted additions and deletions outperform OSEBX in an enhanced index portfolio?"

5.1.1 Preliminary Machine Learning Models Summary

We apply logistic regression using GLM, and tree boosting using XGBoost. In XGBoost we use a standard benchmark model (XGBoost benchmark). We also use an XGBoost model with conditional PPT (XGBoost custom). For each index rebalancing event, we predict either 30, 60, or 100 days prior to ED, all of which happens *before* AD. This is indicated in the suffix of the model names, as summarised in [Table 5.1](#). In total, we are analysing nine unique models. All hyperparameters used in the XGBoost-models are summaries in [Appendix F](#).

Model	ED-30	ED-60	ED-100
GLM	GLM30	GLM60	GLM100
XGBoost Benchmark	XGBB30	XGBB60	XGBB100
XGBoost Custom	XGBC30	XGBC60	XGBC100

Table 5.1: ML Model Names

Overview of ML-models used. We predict using logistic regression (GLM), XGBoost benchmark (XGBB) and XGBoost with conditional PPT (XGBC). For each of these models, we predict 30, 60, and 100 days before ED, which correspond to the model names.

5.1.2 Posterior Probability Threshold Applied

To determine which PPT to apply, we experimented with different thresholds on the models to better understand the direction of the accuracy when changing the threshold. The models seem to be pruned towards predicting more FP and TP classifications compared to FN and TN. Our models are able to correctly predict more additions than deletions, which according to [Figure 1.1](#) might be sub-optimal. [Figure 1.1](#) illustrate that taking short positions on deletions might have higher expected returns than taking long positions on additions. Therefore, we tweaked the PPT slightly upwards. This gave us higher overall accuracy and a more evenly distributed allocation between additions and deletions. In [Table 5.2](#) we show the tweaking of PPT compared to the predictive accuracy for the model GLM30. Here, we choose to illustrate PPT for GLM30, but in general, the other models show a similar pattern as shown for the GLM30 model.

Threshold	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Accuracy	0.9394	0.9464	0.9471	0.9482	0.9452	0.9407	0.9149

Table 5.2: Tuning PPT for ML models

This illustrates the accuracy in relation to PPT for the GLM30 model. We find this model to be somewhat representative, and choose to set PPT for all models to 0.6.

We therefore continue using 0.6 as our initial PPT in the models, but we are yet to determine the ε parameter to be applied in XGBoost custom models (where we use the conditional PPT). To understand how the number of predicted changes behave when changing the ε parameter, we illustrated this relationship in [Figure 5.1](#), fitted to an XGBoost model making predictions 30 days in advance ED. We point out that the total number of predicted changes is the sum of all predicted changes from 2010 and onward.

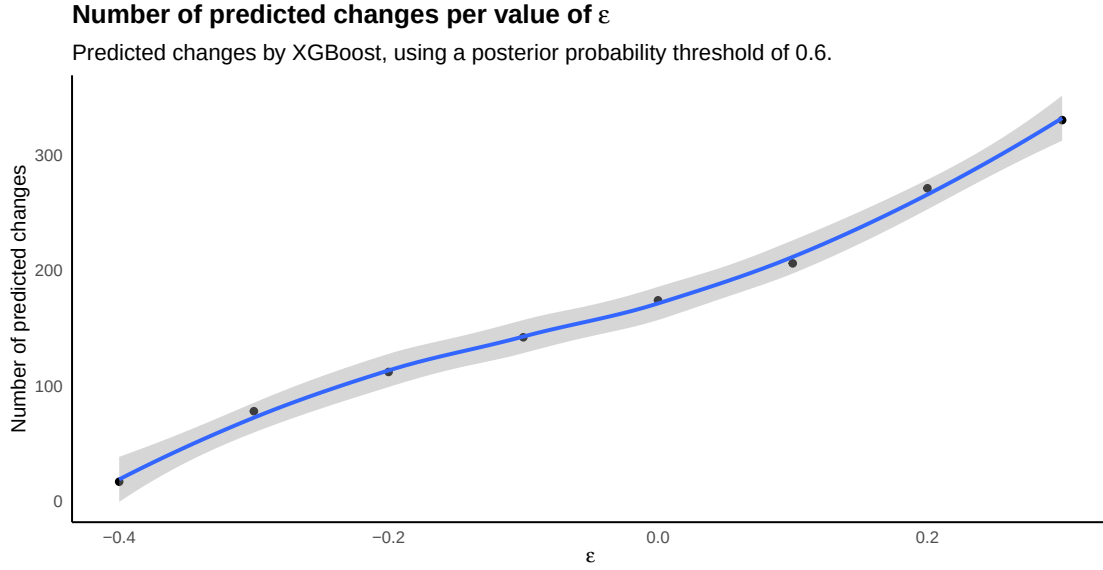


Figure 5.1: Number of Predicted Changes for Values of ε

Sum of predicted additions and deletions as a function of ε in the interval -0.4 to 0.3. Higher ε correspond to increasing the total number of predicted changes. ε represents the parameter added or subtracted from the initial PPT, which is fitted to 0.6. In this plot we used an XGBoost model, predicting 30 days ahead of ED.

Figure 5.1 shows the interval of ε between -0.4 and 0.3. If we had gone with an even smaller ε , to say -0.6, we would not have gotten any predictions at all. Vice versa, if we had used a ε of 0.4, we would have gotten ≈ 2000 predicted changes. We discarded those cases from the plot to make it easier to read, and because they are irrelevant to our application.

To decide which ε to use going forward, we considered different aspects. First of all, we point out that our latter two research questions seek to find out to what degree we can exploit the index effect. The idea of using a conditional PPT is to lower the systematic financial risk and increase the likelihood of buying/selling companies that will be included/excluded from OSEBX. Therefore, we are interested in using a ε that is higher than 0, because ultimately, we want to increase the number of trades. When ε is 0, we get the same predictions for XGBoost custom as the XGBoost benchmark model gives, which is a total of about 170 changes. By increasing ε to 0.2 we predict ≈ 270 changes, meaning that the portfolios will be significantly more diversified and eliminating more systematic risk than the portfolios using XGB benchmark and GLM. As we would

still like to keep accuracy at an acceptable level compared to the other models, we do not aim to extend the ε too far and proceed using 0.2 as our ε parameter in the custom XGBoost models. We would also like to point out that we could have implemented a similar conditional PPT for the GLM models, but we decided to proceed by applying it to the XGBoost models only because they initially are more pruned towards predicting changes than GLM. In practice, the GLM models would require much higher values of ε , and it proved to be more difficult to fit. To conclude this discussion, we proceed using 0.6 as our initial PPT for GLM and XGBB models, and $0.6 +/ - 0.2$ as the conditional PPT for XGBC models.

5.2 Research Question (1): To what degree can machine learning algorithms predict the index composition of OSEBX in the next period?

5.2.1 Model Accuracy

In [Table 5.3](#) we have illustrated the confusion matrices for the classifications for each model in [Table 5.1](#).

		GLM	XGBB	XGBC
$\begin{bmatrix} TN & FN \\ FP & TP \end{bmatrix}$	30	$\begin{bmatrix} 2729 & 117 \\ 112 & 1462 \end{bmatrix}$	$\begin{bmatrix} 2683 & 108 \\ 158 & 1471 \end{bmatrix}$	$\begin{bmatrix} 2639 & 125 \\ 202 & 1454 \end{bmatrix}$
	60	$\begin{bmatrix} 2711 & 111 \\ 110 & 1458 \end{bmatrix}$	$\begin{bmatrix} 2666 & 105 \\ 155 & 1464 \end{bmatrix}$	$\begin{bmatrix} 2628 & 107 \\ 193 & 1462 \end{bmatrix}$
	100	$\begin{bmatrix} 2698 & 111 \\ 105 & 1457 \end{bmatrix}$	$\begin{bmatrix} 2646 & 114 \\ 157 & 1454 \end{bmatrix}$	$\begin{bmatrix} 2605 & 123 \\ 198 & 1445 \end{bmatrix}$

Table 5.3: Confusion matrices for all models

The figure shows the confusion matrices for the different models. TN = true negative (we predicted the company not be included on OSEBX, and it did not), FN = false negative (we predicted the company not be included on OSEBX, but it did) FP = false positive (we predicted the company to be included on OSEBX, but it did not) and TP = true positive (we predicted the company to be included on OSEBX, and it did). The matrices show the sum of all rebalancing events since 2010.

First of all, we notice that the models are predicting more TNs than TPs. This is of course as expected, as only 50-80 of the ≈ 200 companies on OSEAX are also included on OSEBX. More interestingly, we observe quite evenly matched allocations between FP and

FN classifications for the GLM models, meaning that those models are relatively equal in falsely predicting `index_cp = 1` and falsely predicting `index_cp = 0`. For the XGBoost models, there are still some imbalances when it comes to allocation between FP and FN, meaning that XGBoost-models are pruned towards predicting more falsely `index_cp = 1` than `index_cp = 0`. This can mean that an even higher PPT would even those out, but as our tuning of the PPT showed, that can ultimately decrease overall accuracy. Further, the statistics from the confusion matrices are shown in [Table 5.4](#).

Model	Accuracy	Precision	Sensitivity	Specificity	F-score	AUROC	n
<code>index_lag1</code>	0.9507	0.9208	0.9399	0.9565	0.9303	0.9446	0
GLM100	0.9506	0.9292	0.9328	0.9605	0.9310	0.9789	63
GLM60	0.9497	0.9293	0.9298	0.9607	0.9296	0.9808	76
GLM30	0.9482	0.9259	0.9288	0.9589	0.9274	0.9797	93
XGBB60	0.9408	0.9331	0.9043	0.9621	0.9184	0.9817	165
XGBB30	0.9398	0.9316	0.9030	0.9613	0.9171	0.9812	174
XGBB100	0.9380	0.9273	0.9025	0.9587	0.9148	0.9789	168
XGBC60	0.9317	0.9318	0.8834	0.9609	0.9069	0.9817	247
XGBC100	0.9266	0.9216	0.8795	0.9549	0.9000	0.9789	242
XGBC30	0.9260	0.9208	0.8780	0.9548	0.8989	0.9812	271

Table 5.4: Model Accuracy

Model name, accuracy, precision, sensitivity, F-score, AUROC, and the number of predicted changes to index composition (n). The model `index_lag1` is used as a benchmark model, where the prediction is equal to `index_lag1`.

[Table 5.4](#) show the accuracy, precision, sensitivity, specificity, F-score, AUROC, and the number of predicted changes. In addition, we show a simple model named `index_lag1`, where the prediction for the target variable `index_cp` is equal to `index_lag1`. On accuracy, we notice that the ML models were not able to outperform the simple model. This suggests that `index_lag1` is a highly important variable when it comes to predicting index composition. Interestingly, the GLM models were able to outperform the XGBB in terms of accuracy, but we notice that the XGBB and XGBC models generally yield higher AUROC.

Another interesting observation is that GLM100 has higher accuracy than both GLM60 and GLM30, which seems surprising given that GLM100 makes predictions several months ahead of the rebalancing. One possible explanation is that GLM100 predicts so far ahead of the next rebalancing that the date of prediction is closer to the previous rebalancing

than the next. One could therefore possibly expect GLM100 to predict the previous rebalancing with higher accuracy than the upcoming, which in turn can yield almost the same results as the `index_lag1` variable, which seems to be severely important when predicting upcoming composition.

Furthermore, we point out that the number of predicted additions and deletions is higher for XGBoost models compared to GLM, with XGBC30 predicting the highest number of changes. A higher number of trades should in turn lower the financial systematic risk. A *significant insight* from assessing model accuracy is that accuracy seems to diminish with a higher number of predicted changes. To further illustrate this point: the simple model (`index_lag1`) has better accuracy than any of the other nine models but does not predict any changes to the index composition. When trying to exploit the index effect, such a model would therefore not be viable, and we must accept lower accuracy if we want to exploit the index effect.

5.2.2 AUROC

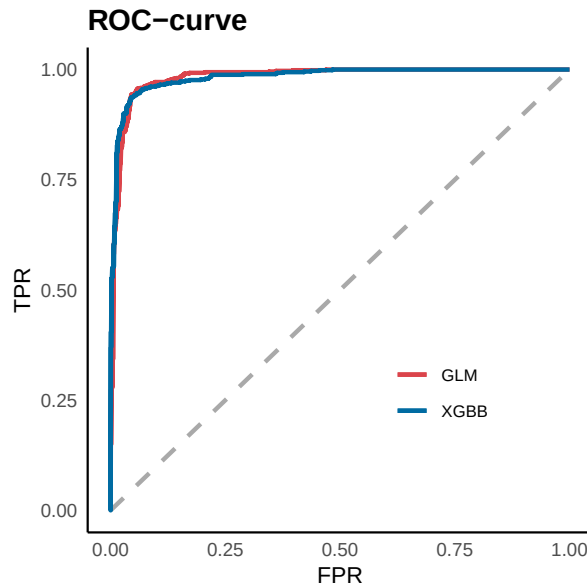


Figure 5.2: ROC Curve for 30-day predictions

Receiver operating characteristic (ROC) curve illustrated for the GLM30 and XGBB30 ML models. Area Under ROC is 0.9797 and 0.9812 for the GLM30 and XGBB30 respectively.

Figure 5.2 show the ROC curve for models GLM and XGBB, for the 30-day prediction horizon. Compared to Figure 3.5, the characteristics from the ROC-curve in Figure 5.2 seem to further support the models' overall ability to predict index-composition, supporting

the findings from Table 5.4. Notice that we did not include the XGBC models in this plot, because the plot is generated by using the estimated likelihoods from the models, which is the same for both XGBB and XGBC models.

5.2.3 Accuracy over Time

Furthermore, we illustrate the models' performance over time, as we are dealing with a time series. Figure 5.3 show the ML model's accuracy over time, completed with the number of changes (additions and deletions) predicted for each period, and the number of true changes captured. As expected from Table 5.4, GLM and XGBB models seem to yield better accuracy for most periods, while predicting fewer changes than XGBC. Once again our results suggest a negative relationship between accuracy and the number of predicted changes.

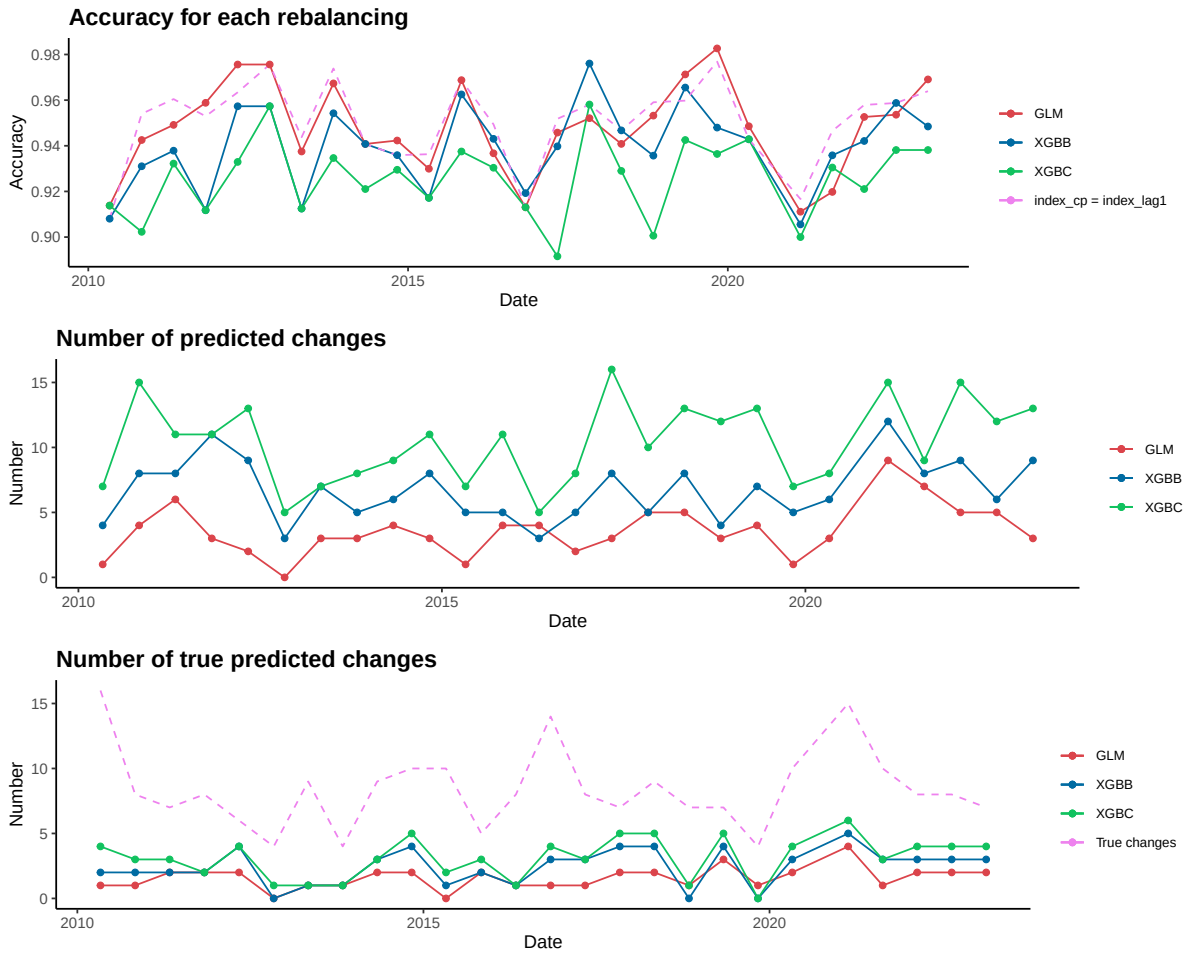


Figure 5.3: ML Results over time, 30-day prediction horizon

Illustrations of how the models GLM30, XGBB30, and XGBC30 perform over the course of the period from 2010 to 2023. Please note that the simple model (index_lag1) is illustrated in the accuracy plot.

Another insight from [Figure 5.3](#) is that we are nowhere near capturing all true changes. As mentioned earlier, if we had bought all companies outside of OSEBX and sold all companies at OSEBX, we would have captured all the changes. However, that would significantly dilute the index effect, and we therefore need to keep accuracy at a reasonable level even for the XGBC models. From these results, we do however learn that the XGBC models are able to capture more of the true changes than the GLM and XGBB models.

5.2.4 Variable Importance Plot

We do already understand that the `index_lag1` variable is important when predicting index composition, but we have yet to explore the importance of the remainder of the variables. [Figure 5.4](#) show variable importance for our GLM- and XGBoost models predicting 30 days prior to ED respectively.

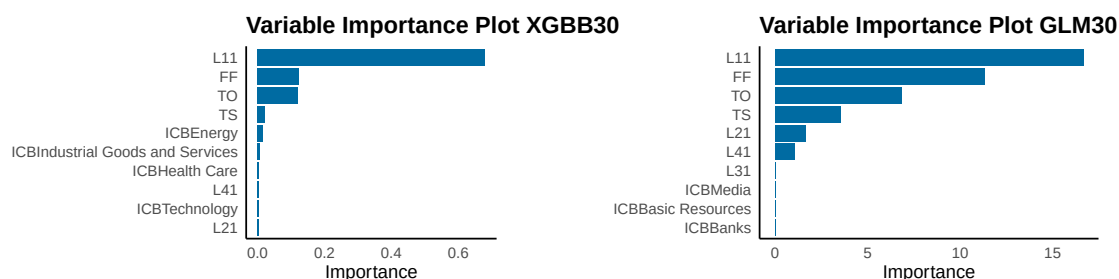


Figure 5.4: VIP for XGBB30 and GLM30

Please see [Table 4.1](#) for explanation for all variable abbreviations. Notice that the x-axis measurement differs between the two plots and each plot must be read independently.

For GLM and XGBB, the variable `index_lag1` proves to be the most important one, followed by `free_float`, `turnover_eob`, and `trading_status`. Understanding why `index_lag1` is most important, we remember that most of the 50-80 companies on OSEBX stay from one period to the next. This includes large Norwegian companies, such as Equinor and Telenor. Therefore, predicting large and liquid companies staying on OSEBX from one period to the next is quite easy, with similar intuition for small listed companies staying off OSEBX. The `index_lag1` variable is however not suited for explaining the cases where the model predicts a change in the response variable compared to the previous period. This must then be explained by the other variables, seemingly by `free_float`, `turnover_eob`, and `trading_status`, which are the variables taken directly

out of the OSEBX rule book. Furthermore, the two models differentiate when it comes to the 5th, 6th, and 7th most important variables. The GLM model seems to depend more on the other lag variables, whereas the XGBoost models are more dependent on the `icb_supersector` variables.

5.2.5 Conclusion for Research Question (1)

In our first research question, we ask "To what degree can ML predict index composition in the next period?". In this section, we have discussed the models' overall performance. The simple model which only predicts using `index_lag1`, gave the highest accuracy (0.9507), but GLM100 followed closely with an accuracy of 0.9506. This means that we were able to predict more than 95% of all observations correctly, and we conclude that ML algorithms can predict the upcoming index composition on OSEBX with high accuracy. Furthermore, we have in this section learned that there might exist a trade-off between model accuracy and the total number of predicted changes made by the ML algorithms. This finding is a *central* insight in our thesis. This can cause great consequences in the following section where we simulate portfolios based on the predicted additions and deletions obtained by the models.

5.3 Research Question (2): To what degree can trading on predicted additions and deletions outperform OSEBX in an active trading portfolio?

In the previous section, we investigated the model accuracy of GLM and XGBoost. In this section, we simulate a portfolio in the period from 2010 to 2023, where we trade on the predicted additions and deletions made by our models. In the portfolio simulation, we consider the number of rebalancing events to predict. We have considered the data available, and find it is suitable for the first rebalancing period to start in 2010. In other words, the first train data set is from 2006 until the first test period in May of 2010. The portfolio simulation ended after the last rebalancing event of March 2023.

5.3.1 Active Trading Strategy

Our trading strategy can be summarised as the following: we buy the companies that the ML models predict to enter OSEBX, and we sell the companies that the ML models predict to exit. The additions/deletions are bought/sold on the date of prediction, and sold/bought at ED. This has several implications. First, we are only allowed to buy companies that are not included in the index at the time of prediction, and we are only allowed to sell companies that currently are on the index. Second, it requires us to expand the strategy to manage the portfolios outside of the active trading period (outside the trading window between prediction and ED). As we want to beat OSEBX, we have decided to take a long position in OSEBX when outside of the trading window. This means that any potential excess returns must come from the active trading period. Thirdly, an important implication is that by using this strategy, we will *never* trade against the index effect in the active period. This is because we are not allowed to buy companies exiting, and not allowed to sell companies entering.

In a long position, one expects the stock price to rise, and in a short position, one expects the stock price to fall (Jacobs et al., 1999). Looking at the index effect in [Figure 1.1](#), we would like to capture both the rising stock prices by going long on additions and falling stock prices by going short on deletions. In practice, this includes finding combinations where `index_lag1=0` is paired with the predicted `index_cp=1` (addition, we take a long position), or where `index_lag1=1` is paired with the predicted `index_cp=0` (deletion, we take a short position). In practice, we simulate two implementations of our strategy. The first strategy includes going long-short, meaning that we are allowed to go both long and short. On the contrary, our second strategy uses long-only positions. The portfolios we simulate and analyse can be summaries as in [Table 5.5](#).

Model	Strategy	ED-30	ED-60	ED-100
GLM	Long-short	GLM30	GLM60	GLM100
XGBoost Benchmark	Long-short	XGBB30	XGBB60	XGBB100
XGBoost Custom	Long-short	XGBC30	XGBC60	XGBC100
GLM	Long-only	GLM30L	GLM60L	GLM100L
XGBoost Benchmark	Long-only	XGBB30L	XGBB60L	XGBB100L
XGBoost Custom	Long-only	XGBC30L	XGBC60L	XGBC100L

Table 5.5: Simulated Portfolio Names

Overview of the portfolio names. Notice that the portfolios trade on predicted changes made by the corresponding model from [Table 5.1](#). The long-short portfolios trade using both predicted additions and deletions. The long-only portfolios trade using only the predicted additions. Long-only portfolios are denoted by the suffix L at the end of the name.

As mentioned in [section 1](#), Euronext rebalances OSEBX twice each year, which is done on ED. Since 2020, ED has been the third Friday of March and September, and before 2020, ED was the first trading day of June and December. Depending on the prediction horizon of 30, 60, or 100 days prior to ED, the active portfolio will be significantly shorter than a year. [Figure 5.5](#) illustrates the timeline of the active trading portfolio investment horizon.

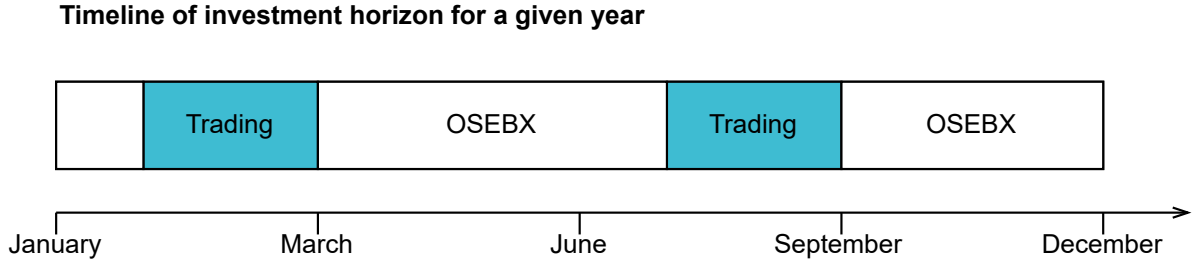


Figure 5.5: Timeline of Investment Horizon

During the year, our portfolios can be in two states. (1) in an active trading period, between the prediction and the ED. (2) in a passive period, meaning that the money is invested in OSEBX.

5.3.2 Financial Assumptions

Next, we discuss the underlying financial assumptions of our portfolio simulation. These are important to make the results robust and practically applicable. The assumptions can be summarised as (1) transaction costs, (2) liquidity, (3) portfolio weights, and (4) interest rate on short positions.

Transaction costs include the brokerage fees when completing a trade (Atkinson et al., 1997). This can come from either brokers implementing the trade by finding a counterpart

directly or using a highly advanced trading algorithm supplied by a broker. Electronic trading algorithms typically have lower transaction costs and can utilise strategies, such as completing trades using VWAP. In the portfolio simulation, we take into account the transaction cost by subtracting a cost per trade made. We use basis points (bps) when discussing trading cost, where 1 bps is equal to 0.01%. NBIM (2020) report commission rates between 5–10 bps during the last two decades, with about 2 bps for electronic trades. DNB (2023) list a fee of 4 bps per transaction for large private investors. In the portfolio simulation, we do electronic trades using VWAP, and assuming we can trade on the terms for large investors according to DNB (2023), we assume a transaction cost of 4 bps.

Liquidity refers to the availability to which stocks can be traded without significantly altering their prices (Hicks, 1962). In practice, stocks on OSEBX may not be available to trade at every given point in time. Still, we conclude that assuming full liquidity for our stocks can be justified. We will be trading stocks on OSEBX that either are on the index or are likely to be on it for the next period. Stocks of these types will have the characteristics of high `trading_status`, representing high liquidity. Using this assumption, we say that all stocks are available, and trades are done using VWAP.

For predicted trades made by the ML model, we must determine the weighting of each security in our active trading portfolio (Frost & Savarino, 1988). We considered different approaches, either to (1) give stocks weights based on the estimated likelihood from the ML model, or to (2) give stocks equal weights. In a practical setting one could argue that weighting smaller or larger positions based on the statistical likelihood is an effective approach, although we decide to weight the positions equally. In short, this is due to ease of implementation, and a lack of an effective converter from probability to weight.

When taking short positions, one has to borrow a number of stocks from a third party (Jacobs et al., 1999). This comes with an interest rate cost, as the original stock owner is compensated for giving up the opportunity to trade during the lending period. The short-position interest rate depends on a variety of factors, including supply of short-sellers. Nordnet (2023) state a short-position interest rate of 4% annually, and this is what we use in our simulations. Furthermore, regarding short positions, we have used a maximum limit of what the short position might rise to before we are forced to buy back the position. This limit is 100 %, meaning that if the stock price doubles in value, then we are forced

to buy it back and we will get a return of -100%. In other words, this diminishes the loss, to the maximum loss in a long position.

To sum up, our financial assumptions aim to be conservative, but yet insight-full as shown in [Table 5.6](#). This is to increase the robustness of our findings and make them valid in a practical setting.

Financial	Assumption
Transaction cost trades	4 bps
Transaction cost OSEBX	4 bps
Liquidity	Full
Price	VWAP
Stock weights in portfolio	Equal
Risk-free rate R_f	$\approx 1.46\%$ annually
Short-position interest rate	4% annually

Table 5.6: Summary of Financial Assumptions in the Trading Portfolios

These assumptions lay the basis for our simulations. They aim to be as accurate in a practical sense as possible.

5.3.3 Application of the Portfolio Simulations

Before presenting portfolio performance, we present a brief explanation to how the simulations are executed. First, the ML models generate a set of predictions, for 30-, 60-, and 100-days in advance of the ED. The data frame in which the predictions are stored is then filtered, so that only the cases where the prediction is different from the classification in the previous period are left. Next, the VWAP is extracted for each recommended trade on both ED and on the date of prediction, depending on the prediction horizon. The return of the position is then determined as shown in [Equation 5.1](#), where r_l is return on long-positions, and r_s is return on short-positions, p_0 is the VWAP of the stock at the start of the trading period, and p_1 is the VWAP at the end.

$$r_l = \frac{p_1 - p_0}{p_0}, \quad r_s = \frac{p_0 - p_1}{p_0} \quad (5.1)$$

Next, we take the average of each return per trading period (because we assume equal weighting in our portfolios), and use this return as the period return. This operation is done for each trading period, and we are left with the portfolio development over time. The amount of money that is held at the beginning of the trading period is then multiplied

by the average growth factor (the average return plus 1). The same logic applies to the passive periods, where the money is invested in OSEBX. Transaction costs are applied at the start and end of each trading period, and the short interest rate is subtracted from the calculated return on any short position. This is explained in greater detail in [Appendix I](#). In [Appendix J](#) we also show an example of a simulated portfolio, namely the GLM100 portfolio.

5.3.4 Performance in a Capital Asset Pricing Model

The financial metrics for our respective active trading portfolios are shown in [Table 5.7](#), sorted by decreasing SR. The results in [Table 5.7](#) can be compared to the desired metrics in [Table 2.1](#). All calculations are done using monthly returns, except TE_{ann} . Notice that the names of portfolios in [Table 5.7](#) correspond to the ML model used to generate trade predictions, as shown in [Table 5.5](#).

Portfolio	$R_p - R_b$	R_p	σ_p	SR	α_{CAPM}	β	TE_{ann}	IR
OSEBX	0	0.0095	0.0423	0.1981	0.0002	1.0148	0.0000	
GLM60	0.0095	0.0190	0.0756	0.2367	0.0139	0.4966	0.2599	0.1268
GLM30	0.0034	0.0129	0.0511	0.2309	0.0048	0.8634	0.1260	0.0940
XGBC30	0.0017	0.0112	0.0441	0.2289	0.0026	0.9249	0.0789	0.0755
GLM100	0.0087	0.0182	0.0757	0.2257	0.0133	0.4684	0.2668	0.1132
XGBB30	0.0010	0.0105	0.0488	0.1917	0.0016	0.9536	0.1056	0.0321
XGBC60	-0.0029	0.0066	0.0480	0.1138	-0.0004	0.7285	0.1383	-0.0731
XGBB60	-0.0019	0.0076	0.0608	0.1073	0.0020	0.5532	0.2095	-0.0306
XGBC100	-0.0038	0.0057	0.0443	0.1027	0.0017	0.3470	0.1782	-0.0744
XGBB100	-0.0058	0.0037	0.0539	0.0487	-0.0012	0.4696	0.1947	-0.1024
XGBC30L	0.0032	0.0127	0.0482	0.2398	0.0034	1.0065	0.0903	0.1222
XGBB30L	0.0033	0.0128	0.0500	0.2335	0.0033	1.0386	0.0990	0.1158
GLM30L	0.0009	0.0104	0.0460	0.2011	0.0012	0.9942	0.0674	0.0450
XGBB60L	0.0011	0.0106	0.0543	0.1751	0.0019	0.9428	0.1361	0.0288
XGBC60L	0.0004	0.0099	0.0505	0.1737	0.0008	0.9843	0.1063	0.0131
GLM60L	0.0002	0.0097	0.0553	0.1550	0.0002	1.0340	0.1187	0.0056
XGBC100L	0.0011	0.0106	0.0649	0.1468	0.0035	0.7391	0.2078	0.0191
XGBB100L	0.0015	0.0110	0.0682	0.146	0.00324	0.826	0.211	0.0255
GLM100L	-0.0031	0.0064	0.0669	0.0797	-0.0033	1.0614	0.1809	-0.0583

Table 5.7: CAPM Portfolio Performance for Active Trading Strategies Using Monthly Returns

Risk free interest rate is $\approx 0.12\%$ per month. The market return is assumed to be the return of OSEAX during the same period, which gave a monthly average return of $\approx 0.92\%$. Both the risk-free interest rate and the market return are calculated based on (Ødegaard, 2023a) in the period from January 31st, 2010 to May 31st, 2022. TE and information ratio is calculated in relation to OSEBX, as this is the index we are aiming to beat. The table is sorted after SR. The analysed period is from January 31st, 2010 to May 31st, 2022. R_b represents the return of the benchmark index OSEBX, so the first column shows excess returns in relation to OSEBX.

Please note that we have defined the market to be OSEAX, as this is the pool of stocks we can buy or sell. Our first observation is that several portfolios yield higher returns R_p than OSEBX. Especially the GLM portfolios yield exceptional returns (GLM60 yielding twice the return of OSEBX). However, those portfolios tend to also be involved with higher risk σ_p . Taking this into consideration, we observe SR, where the XGBC30L portfolio excels. It is, however, closely followed by the portfolios GLM60, GLM30, XGBC30, GLM100 and XGBB30L. All of these portfolios outperform OSEBX in terms of SR, which seems promising. Furthermore, we observe the biggest alphas α_{CAPM} in GLM60 and GLM100. The high α_{CAPM} for those two portfolios is likely caused by the low correlation with the market (β) and exceptional returns. At this stage, we would expect this α_{CAPM} to diminish after adjusting for FF3 later on. We also observe notable α_{CAPM} for GLM30, XGBC30, XGBC30L and XGBB30L. Furthermore, the calculated betas β seems somewhat surprising, considering the risk σ_p involved with several of the portfolios. In general, a risky portfolio is considered to have a high beta (namely, greater than 1). In the long-short portfolios we do however trade on short-positions, which in the active trading periods can constitute a negative correlation with the market, and in turn, lower the beta β (being the correlation coefficient with the market). When considering the long-only portfolios we observe somewhat more expected results, with β varying from 0.73 to 1.04. In conclusion, several models outperform OSEBX in the world of CAPM, both on R_p , SR , and α_{CAPM} . Further, we investigate the portfolio's exposure to risk.

5.3.5 Fama-French 3 factor Model on Active Portfolios

Going further than CAPM portfolio performance, we want to understand if the excess returns can be explained by the risk factors in an FF3 model. We run a linear regression using portfolio returns minus risk-free rate as the dependent variable, and factors SMB, HML, and the market risk premium (see [section 2](#)) as independent variables. The FF3 factors introduced in [section 2](#) are retrieved from Ødegaard ([2023a](#)), as discussed in [section 4](#).

Note that Ødegaard ([2023a](#)) report Fama-French factors for OSE, which is somewhat different from OSEBX. Because OSEAX represents more of the companies on OSE than OSEBX does, we argue that the risk factors are representing OSEAX to a higher extent than OSEBX. We therefore use OSEAX as the market in the FF3 regressions. We have

also done the same regressions using OSEBX as the market, which is shown in [Appendix K](#), to check for any significant differences.

	Dependent variable:									
	GLM			XGBB			XGBC			
	30	60	100	30	60	100	30	60	100	OSEBX
MR	0.890*** (0.075)	0.553*** (0.149)	0.510*** (0.151)	0.958*** (0.060)	0.545*** (0.117)	0.457*** (0.104)	0.938*** (0.046)	0.729*** (0.078)	0.337*** (0.087)	1.022*** (0.017)
SMB	−0.166** (0.076)	−0.133 (0.150)	−0.129 (0.152)	−0.105* (0.061)	−0.068 (0.117)	0.138 (0.104)	−0.124*** (0.046)	−0.058 (0.078)	0.101 (0.087)	−0.042** (0.017)
HML	−0.055 (0.056)	−0.219** (0.110)	−0.153 (0.112)	0.029 (0.045)	0.070 (0.086)	−0.005 (0.077)	−0.008 (0.034)	0.023 (0.058)	0.007 (0.064)	−0.016 (0.013)
α_{FF3}	0.006** (0.003)	0.013** (0.006)	0.013** (0.007)	0.003 (0.003)	0.004 (0.005)	−0.003 (0.004)	0.004** (0.002)	0.001 (0.003)	0.0004 (0.004)	0.001 (0.001)
Obs	148	148	148	148	148	148	148	148	148	148
R ²	0.494	0.099	0.078	0.647	0.145	0.138	0.746	0.388	0.112	0.962
R ² _{adj}	0.484	0.080	0.059	0.639	0.127	0.120	0.740	0.375	0.093	0.961

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 5.8: FF3 Regression: Long-Short Active Portfolios

The table shows the regression summary of the long-short portfolios return net of risk-free interest rate against market risk premium (RM), the small minus big factor (SMB), and the high minus low factor (HML). The market is considered to be OSEAX, the same as we used as the market used in [Table 5.7](#). Notice that the factors are reported for OSE, and was obtained by Ødegaard ([2023a](#)). We, therefore, decided to use OSEAX as the market, as this more closely represents the market that the factors express.

In [Table 5.8](#), we find statistically significant positive alpha α_{FF3} at a 95% confidence level for portfolios GLM30, GLM60, GLM100, and XGBC30, all outperforming the benchmark index OSEBX α_{FF3} of 0.001. This indicates that the mentioned portfolios produce excess returns that are not fully explained by FF3. In comparison to the findings in [Table 5.7](#), we see that the GLM60 and GLM100 portfolios diminish in FF3-regression (compared to the α_{CAPM} reported in CAPM), whereas both GLM30 and XGBC30 increases slightly. This can suggest that the GLM30 and XGBC30 have negative exposure to the risk factors. This is further supported by the correlation coefficients for both SMB and HML. There might also be a source of error coming from the difference between what we have defined as the market, OSEAX, and OSE, on which the risk factors are based.

	<i>Dependent variable:</i>									OSEBX
	GLML			XGBBL			XGBCL			
	30	60	100	30	60	100	30	60	100	
MR	1.001*** (0.045)	1.042*** (0.074)	1.066*** (0.104)	1.030*** (0.055)	0.936*** (0.079)	0.795*** (0.025)	1.006*** (0.052)	0.984*** (0.064)	0.719*** (0.118)	1.022*** (0.017)
SMB	−0.072 (0.045)	0.026 (0.074)	0.143 (0.105)	0.003 (0.055)	0.055 (0.080)	0.264 (0.025)	−0.007 (0.053)	0.032 (0.064)	0.170 (0.119)	−0.042** (0.017)
HML	0.001 (0.033)	−0.056 (0.054)	−0.098 (0.077)	0.038 (0.041)	0.005 (0.059)	0.017 (0.018)	0.002 (0.039)	−0.020 (0.047)	0.024 (0.088)	−0.016 (0.013)
α_{FF3}	0.002 (0.002)	−0.001 (0.003)	−0.006 (0.005)	0.004 (0.002)	0.001 (0.003)	−0.001 (0.001)	0.004 (0.002)	0.0002 (0.003)	0.001 (0.005)	0.001 (0.001)
Obs	148	148	148	148	148	148	148	148	148	148
R ²	0.783	0.588	0.437	0.720	0.505	0.189	0.727	0.634	0.229	0.962
R ² _{adj}	0.778	0.579	0.425	0.714	0.494	0.172	0.721	0.626	0.213	0.961

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 5.9: FF3 Regression: Long-Only Active Portfolios

The table shows the regression summary of the long-only portfolios' return net of risk-free interest rate against market risk premium (RM), the small minus big factor (SMB), and the high minus low factor (HML). The market is considered to be OSEAX, similar to the market used in Table 5.7. Notice that the factors are reported for OSE, and was obtained by Ødegaard (2023a)

Looking at Table 5.9, we do not get the same alphas as we did for the long-short portfolios. In the long-only portfolios, most portfolios are positively correlated with the risk factors (but not on a significant level). This might explain why we are unable to get significant alphas α_{FF3} in the long-only portfolios. We also see that most alphas decrease in contrast to the calculated alphas in Table 5.7, indicating that the long-only portfolios are relatively exposed to the risk factors.

5.3.6 Plot of Portfolios

Next, we plot the portfolios with their cumulative return to get a more visual representation of their performance. The long-short active portfolios are shown in Figure 5.6. This is the GLM30, GLM60, GLM100 and XGBC30. The corresponding long-only portfolios are plotted in the in Figure 5.7. Notice that we have only plotted the portfolios yielding a significant alpha in FF3. All portfolios are plotted in Appendix H, in Figure H.9 and Figure H.10. In the same appendix, we also plotted so-called perfect portfolios, which are simulated knowing all additions and deletions, in Figure H.8.

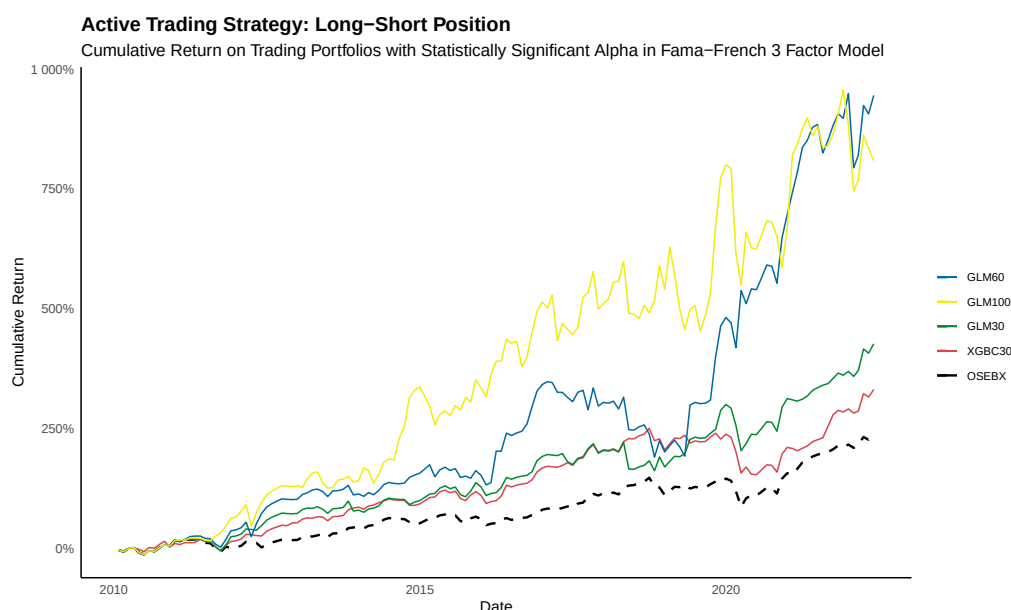


Figure 5.6: Cumulative Return Long-Short Active Trading Strategy

Cumulative return for GLM30, GLM60, GLM100 and XGBC30, which all shows a significant alpha in the FF3-regression. OSEBX is also displayed. The GLM60 and GLM100 portfolios provide great investment opportunity for the risk-seeking investor.

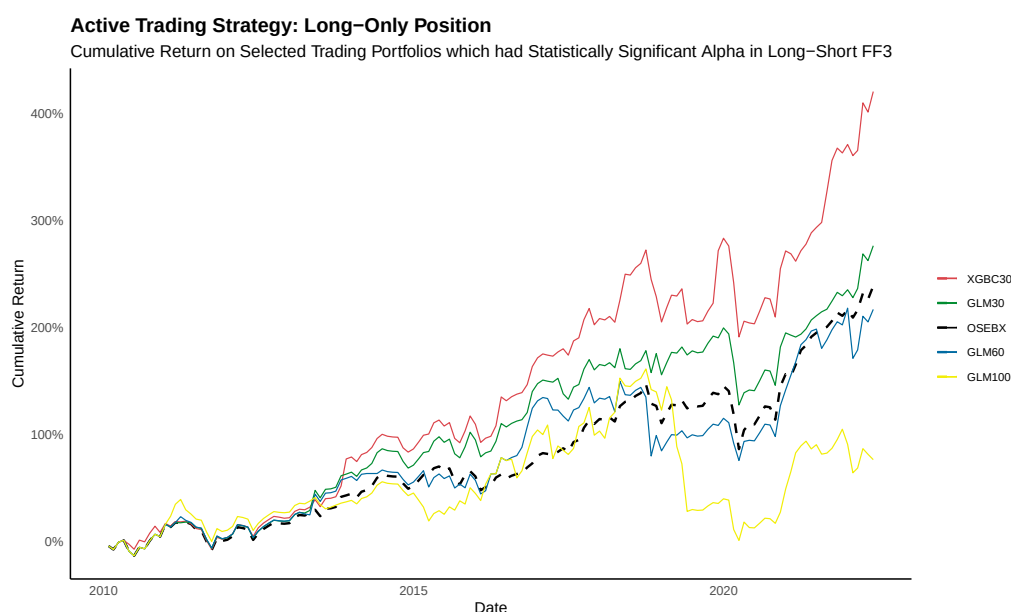


Figure 5.7: Cumulative Return Long-Only Active Trading Strategy

Cumulative return for GLM30, GLM60, GLM100 and XGBC30, which all shows a significant alpha in the FF3-regression. OSEBX is also displayed. Still great returns for XGBC30 and GLM30, not so much for GLM60 and GLM100.

The cumulative return illustrated in [Figure 5.6](#) shows that the GLM60 and GLM100 portfolios are yielding exceptional returns, though severely volatile. Those two portfolios are significantly affected by the low number of shares held in the active trading period,

which varies between 1 and 6 companies. In contrast, the XGBC30 portfolio consist of about five times more, and still made for a significant alpha in the FF3-regression. Considering the long-only portfolios in [Figure 5.7](#), a different pattern emerges, where XGBC30L yields the highest returns, and the GLM60L and GLM100L perform worse than OSEBX when considering cumulative return. The predictions made by model XGBC30 seem to perform well in both strategies, and the increased number of shares seems to lower the risk in practice.

5.3.7 Conclusion for Research Question (2)

In this section, we have presented and discussed the results, to answer the research question: "To what degree can trading on predicted additions and deletions outperform OSEBX in an active trading portfolio?". In this research question, we find that two types of very different portfolios tend to perform well. (1) The GLM portfolios predict few changes, but obtain excellent return on the few trades they do. (2) The XGBC30 portfolio can also obtain excellent returns but with a level of risk matching the benchmark, by predicting more changes than the GLM portfolios. Both directions tend to outperform OSEBX and have shown significant alphas in FF3. This is a *key insight* in exploiting the index effect. Based on the results, it becomes evident that trading on predicted additions and deletions has proven to outperform OSEBX for the last 13 years, and we conclude that this strategy can outperform OSEBX to a high degree.

5.4 Research Question (3): To what degree can trading on predicted additions and deletions outperform OSEBX in an enhanced index portfolio?

In the previous two sections, we assessed the ML model performance of GLM and XGBoost and also applied them in an active trading strategy. Now we want to go further and investigate if such an active trading strategy can be combined with passive index investing to create an enhanced index portfolio, outperforming OSEBX.

5.4.1 Enhanced Index Strategy

The enhanced index strategy seeks to combine the active trading strategy with an otherwise passive index portfolio. As discussed in [section 1](#), defining enhanced index can be difficult (Riepe & Werner, 1998). State Street (2020) report that index investing can be considered "enhanced" with a TE in the continuum from 0.5% to 2%. We have decided to use a TE_{ann} of 2%, which is in the upper range of enhanced index. This enables us to more easily compare portfolio performances (as performance differences will grow with active share and TE).

In practice, we adjust some of the financial assumptions, and run the simulations again, and use those to combine with OSEBX to construct enhanced index portfolios. We illustrate the enhanced index strategy in [Figure 5.8](#). Compared to the active trading strategy in [Figure 5.5](#), we see that the active share is smaller, and the portfolio as a whole can therefore move closer to the benchmark index OSEBX. Note that we still use OSEBX as a benchmark index, which we calculate TE against, while we use OSEAX as the market in financial models.

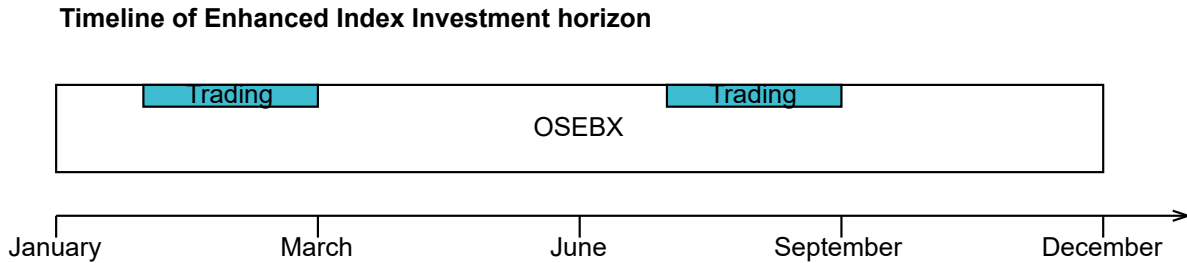


Figure 5.8: Timeline of Enhanced Index Investment horizon

In the enhanced index portfolios, the active trading constitutes only for a smaller part of the total portfolio, which is dominated by OSEBX.

The simulated enhanced index funds consist of two parts. (1) The active share, which is a portfolio similar to those analysed in the previous section, and (2) The passive share, which is invested in OSEBX at all times. In contrast to the active trading perspective discussed in the previous section, we are now forced to own both the passive and the active share at the same time. The return of the enhanced index portfolios can be expressed as shown in [Equation 5.2](#), where r_E is the return on the enhanced index portfolio, r_A is the

return on the active share, r_P is the return of the passive share, and s describes how big the active share is in the enhanced index portfolio.

$$r_E = r_A * s + r_P * (1 - s) \quad (5.2)$$

5.4.2 Financial Assumptions

From the perspective of enhanced index portfolios, we face different challenges than those from the active trading portfolios. In this section, we are going to explore the most significant assumptions regarding the enhanced index portfolio, which can be broken down into interest rate on short positions, and share of active management.

We assume that by using an enhanced index strategy, we are managing a larger enhanced index fund, that follows OSEBX closely. By making this assumption, we always own stocks on OSEBX before each active investment horizon. This changes the assumption regarding short-position interest rates. Most importantly, we do not have to pay interest rates on the short positions. We would simply sell the positions because we already own them in the passive portfolio. Therefore, we simulate the active portfolios again, but this time without using interest rates on any short positions.

Second, we assume that we are not allowed to operate outside the annual TE of 2%. This limits how big a share of active management we can allow for throughout the year. How large the active management can be will depend on the prediction horizon of 30, 60, or 100 days, and also the volatility of the active portfolio. In this thesis, we have maximised the active share, given that TE cannot exceed 2% annually. The weighting of the active share is therefore significantly varied between the portfolios. In general, the active portfolios with higher volatility are paired with a lower active share in the enhanced index portfolios. The weightings are shown in table [Table 5.10](#).

Third, the assumptions made in the previous section also apply in the context of an enhanced index portfolio, namely the assumptions regarding transaction costs, liquidity, and active portfolio weights (equal weighting of the companies held in the active part of the portfolio).

5.4.3 Application of the Enhanced Portfolio Simulations

The practical implementation of our enhanced index portfolio simulation closely mirrors that of the active trading portfolio simulation, but the main difference is that the enhanced portfolio is a combination of the active and the passive share. In practice, we first run new simulations of the active portfolios, without interest rates on short positions. Next, we optimise each active share, to yield a TE of 2%. Finally, we analyse the performance of the enhanced index portfolios which are put together by combining the passive and active share using the weighting presented in [Table 5.10](#).

Enhanced Portfolio	Active Share	Passive Share	TE _{ann}
GLM30L	0.267	0.733	2%
XGBC30	0.221	0.779	2%
XGBC30L	0.182	0.818	2%
GLM60L	0.170	0.830	2%
XGBB30	0.166	0.834	2%
XGBB30L	0.166	0.834	2%
XGBC60L	0.159	0.841	2%
XGBC60	0.141	0.859	2%
XGBB60L	0.126	0.874	2%
GLM100L	0.124	0.876	2%
GLM30	0.123	0.877	2%
XGBC100	0.112	0.888	2%
XGBB100	0.110	0.890	2%
XGBC100L	0.098	0.902	2%
XGBB60	0.080	0.920	2%
XGBB100L	0.066	0.934	2%
GLM60	0.041	0.959	2%
GLM100	0.028	0.972	2%

Table 5.10: Active and Passive Share in Enhanced Index Portfolios

All portfolios are constructed to maximise the active share, and the constraint of TE cannot exceed 2% per year. The active share represents the simulated portfolio trading on predicted additions/deletions, and the passive share is OSEBX.

In [Table 5.10](#), we present the maximum weightings of the active share which is allowed in the enhanced index portfolios. This results in some interesting observations. First, we observe that the two most extreme active portfolios from the previous section (namely GLM60 and GLM100) are only allowed to constitute 4.1% and 2.8% as the active shares respectively. Secondly, the low-risk active portfolios like GLM30L (26.7%), XGBC30 (22.1%), and XGBC30L (18.2%) are allowed to constitute an active share of several times more than the extreme portfolios.

5.4.4 Performance in the Capital Asset Pricing Model

Next, we investigate enhanced index portfolio performance. Table 5.11 illustrate performance for the portfolios, sorted by decreasing SR. Notice that the name of the portfolio in Table 5.11 corresponds to the active portfolio from Table 5.1 used as the active share (without interest rate on short positions).

Portfolio	$R_p - R_b$	R_p	σ_p	SR	α_{CAPM}	β	TE _{ann}	IR
OSEBX	0	0.0095	0.0423	0.1981	0.0002	1.0148	0.0000	
GLM60	0.0005	0.0100	0.0410	0.2180	0.0011	0.9735	0.0200	0.0971
GLM30	0.0005	0.0100	0.0418	0.2117	0.0008	0.9915	0.0200	0.0802
GLM100	0.0003	0.0098	0.0412	0.2112	0.0008	0.9812	0.0200	0.0572
XGBC30	0.0004	0.0099	0.0416	0.2111	0.0008	0.9933	0.0200	0.0721
XGBB30	0.0001	0.0096	0.0420	0.2026	0.0004	1.0046	0.0200	0.0225
XGBC100	-0.0005	0.0090	0.0392	0.2024	0.0003	0.9378	0.0200	-0.0771
XGBB60	-0.0003	0.0092	0.0407	0.2001	0.0003	0.9702	0.0200	-0.0413
XGBC60	-0.0004	0.0091	0.0407	0.1958	0.0001	0.9715	0.0200	-0.0693
XGBB100	-0.0006	0.0089	0.0399	0.1958	0.0001	0.9545	0.0200	-0.0970
XGBC30L	0.0006	0.0101	0.0423	0.2129	0.0008	1.0144	0.0200	0.1105
XGBB30L	0.0006	0.0101	0.0425	0.2112	0.0007	1.0217	0.0200	0.1033
GLM30L	0.0002	0.0097	0.0425	0.2021	0.0004	1.0083	0.0200	0.0359
XGBC100L	0.0000	0.0095	0.0414	0.2015	0.0003	0.9908	0.0200	-0.0057
XGBB60L	0.0000	0.0095	0.0420	0.2008	0.0003	1.0044	0.0200	0.0104
XGBB100L	0.0000	0.0095	0.0420	0.1995	0.0002	1.0034	0.0200	-0.0006
XGBC60L	0.0000	0.0095	0.0423	0.1982	0.0002	1.0092	0.0200	0.0004
GLM60L	-0.0001	0.0094	0.0429	0.1942	0.0001	1.0190	0.0200	-0.0087
GLM100L	-0.0004	0.0091	0.0426	0.1872	-0.0003	1.0180	0.0200	-0.0705

Table 5.11: CAPM Portfolio Performance for Enhanced Index Strategies using Monthly Returns

Risk free interest rate is ≈ 0.12 % per month. The market return is assumed to be the return of OSEAX during the same period, which gave a monthly average return of $\approx 0.92\%$. Both risk free interest rate and market return is calculated based on (Ødegaard, 2023a) in the period from January 31st 2010 to May 31st 2022. The TE and information ratio is calculated in relation to OSEBX, as this is the index we are aiming to beat. The table is sorted after SR. The analysed period is from January 31st 2010 to May 31st 2022.

Looking at Table 5.11, we find that GLM portfolios once again yield high SR, closely followed by XGBC30, XGBC30L, and XGBB30L. Also, when looking at the alpha α_{CAPM} , GLM60 yields the highest, followed by GLM30, GLM100, XGBC30 and XGBC30L. In the enhanced portfolio simulation, the XGBC30L portfolio outperforms OSEBX the most, which seems to be a result of it being allowed to constitute for a significantly larger share (18.2%) than the competing volatile portfolios. As we observed in the analysis of the active portfolios in the previous section, we see a tendency where the two styles of portfolios

seem to outperform OSEBX. The first one is the high-yielding volatile portfolios, such as GLM60 and GLM100, and the second one is the diverse, low-risk portfolios.

5.4.5 Fama-French 3-factor Model

In [Table 5.12](#) and [Table 5.13](#) we show FF3 regressions for long-short and long-only enhanced index portfolios respectively.

	<i>Dependent variable:</i>									
	GLM			XGBB			XGBC			OSEBX
	30	60	100	30	60	100	30	60	100	
MR	1.001*** (0.021)	0.984*** (0.020)	0.992*** (0.020)	1.011*** (0.018)	0.976*** (0.019)	0.959*** (0.018)	1.002*** (0.019)	0.977*** (0.019)	0.943*** (0.017)	1.022*** (0.017)
SMB	−0.060*** (0.021)	−0.046** (0.020)	−0.047** (0.020)	−0.054*** (0.018)	−0.047** (0.019)	−0.027 (0.018)	−0.064*** (0.019)	−0.046** (0.019)	−0.027 (0.017)	−0.042** (0.017)
HML	−0.020 (0.015)	−0.029* (0.015)	−0.031** (0.015)	−0.007 (0.013)	−0.005 (0.014)	−0.008 (0.013)	−0.013 (0.014)	−0.007 (0.014)	−0.010 (0.013)	−0.016 (0.013)
α_{FF3}	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.0004 (0.001)	0.001* (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
Obs	148	148	148	148	148	148	148	148	148	148
R ²	0.943	0.943	0.947	0.957	0.950	0.954	0.952	0.951	0.956	0.962
R ² _{adj}	0.942	0.942	0.946	0.956	0.949	0.953	0.951	0.950	0.955	0.961

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 5.12: FF3 Regression: Long-Short Enhanced Index

The table shows the regression summary of the long-short enhanced portfolios return net of risk-free interest rate against market risk premium (RM), the small minus big factor (SMB), and the high minus low factor (HML). The market is considered to be OSEAX, the same we used in [Table 5.7](#). Notice that the factors are reported for OSE, and were obtained by Ødegaard ([2023a](#))

First of all, we notice that the coefficients have converged when comparing to the active trading FF3-regressions in [Table 5.8](#) and [Table 5.9](#). This most likely comes from the fact that the enhanced index portfolios track OSEBX closely, and all portfolios are forced to track the index within the TE limit of 2%. In other words, they will share risk and return characteristics with the benchmark. Interestingly, most of the coefficients to the risk factors are negative for enhanced index portfolios, similar to the same direction as the coefficients for the active portfolios [Table 5.8](#). A likely explanation for this is that OSEBX itself seems to be negatively correlated with the risk factors. Interestingly, most

of the alphas for the long-short positions seem to be around 0.001. However, only the XGBC30 is significant (at a 90% confidence level).

	<i>Dependent variable:</i>									
	GLML			XGBBL			XGBCL			
	30	60	100	30	60	100	30	60	100	OSEBX
MR	1.015*** (0.021)	1.026*** (0.021)	1.025*** (0.019)	1.025*** (0.017)	1.010*** (0.019)	1.007*** (0.019)	1.020*** (0.018)	1.015*** (0.020)	0.995*** (0.018)	1.022*** (0.017)
SMB	−0.051** (0.022)	−0.030 (0.021)	−0.026 (0.019)	−0.032* (0.017)	−0.028 (0.019)	−0.017 (0.019)	−0.035* (0.018)	−0.030 (0.020)	−0.019 (0.018)	−0.042** (0.017)
HML	−0.010 (0.016)	−0.021 (0.016)	−0.021 (0.014)	−0.003 (0.012)	−0.013 (0.014)	−0.011 (0.014)	−0.010 (0.013)	−0.016 (0.014)	−0.013 (0.014)	−0.016 (0.013)
α_{FF3}	0.001 (0.001)	0.0003 (0.001)	−0.0001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.0004 (0.001)	0.001 (0.001)	0.0004 (0.001)	0.0004 (0.001)	0.001 (0.001)
Obs	148	148	148	148	148	148	148	148	148	148
R ²	0.941	0.943	0.954	0.964	0.953	0.953	0.958	0.950	0.955	0.962
R ² _{adj}	0.940	0.942	0.953	0.963	0.952	0.952	0.957	0.949	0.954	0.961

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 5.13: FF3 Regression: Long-Only Enhanced Index

The table shows the regression summary of the long-only enhanced portfolios return net of risk free interest rate against market risk premium (RM), the small minus big factor (SMB), and the high minus low factor (HML). The market is considered to be OSEAX, the same we used in [Table 5.7](#). Notice that the factors are reported for OSE, and were obtained by Ødegaard ([2023a](#))

Furthermore, when considering the FF3 regressions for the enhanced long-only portfolios we get results quite similar to the enhanced long-short regressions, where the differences between the portfolios have converged. Also, we notice that none of the enhanced long-only portfolios are yielding significant alphas in the FF3-regressions. Even the strongest performing enhanced portfolio measured in R_p XGBC30L is not significant, and neither are the other portfolios with high SR.

5.4.6 Plot of Portfolios

Next, we plot the portfolios with their cumulative return in comparison to the benchmark index OSEBX. The long-short and long-only enhanced index portfolios are shown in [Figure 5.9](#) and [Figure 5.10](#) respectively. The portfolios illustrated here are the same as we showed in the previous section (but now in enhanced portfolios). This is the GLM30,

GLM60, GLM30, and XGBC30. Notice that all simulated enhanced portfolios are shown in [Appendix H](#), in [Figure H.11](#) and [Figure H.12](#).

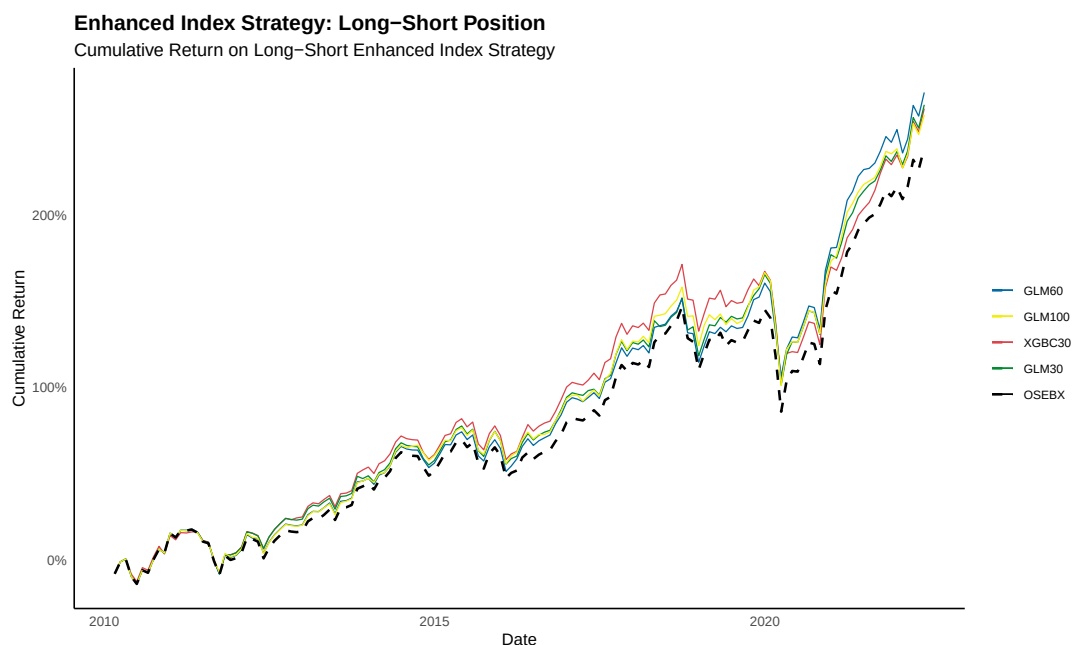


Figure 5.9: Cumulative Return Long–Short Enhanced Index

Cumulative return for the enhanced long-short portfolios GLM30, GLM60, GLM100, XGBC30 and OSEBX. Only the XGBC30 portfolio yields a significant alpha in the FF3-regression.



Figure 5.10: Cumulative Return Long–Only Enhanced Index

Cumulative return for the enhanced long-only portfolios GLM30L, GLM60L, GLM100L, XGBC30L and OSEBX. None of the portfolios yielded significant alphas in the FF3-regression.

When considering the visual representation of cumulative return for enhanced portfolios,

once again a pattern emerges where the two portfolio types seem to perform well over time. In contrast to the active portfolios analysed in the previous section, the enhanced portfolios are seemingly less exposed to volatility, following OSEBX closely.

5.4.7 Conclusion for Research Question (3)

Finally, our third research question was "To what degree can trading on predicted additions and deletions outperform OSEBX in an enhanced index portfolio?". In this section, we have analysed enhanced index portfolios, which were generated by combining the active trading portfolios from the previous section, with a passive share in OSEBX. The enhanced index portfolios were optimised to have a TE_{ann} of 2%. In [Table 5.11](#), we found that several enhanced index portfolios yielded positive CAPM α_{CAPM} and outperformed OSEBX on metrics like R_p and SR. But, in [Table 5.12](#) and [Table 5.13](#), only the XGBC30 enhanced index portfolio gave a significant FF3 α_{FF3} at a 10% confidence level after adjusting for risk factors. As expected, when combining the somewhat extreme GLM60 and GLM100 active portfolios with a passive portfolio, we are forced to use a lower active share due to the TE limit of 2%. In contrast, the risk-averse portfolios allow for a bigger active share, making XGBC30L the highest-yielding enhanced index portfolio.

In the previous section, we found that one can exploit the index effect by trading on (1) a few accurate predictions, or (2) more changes with a level of risk matching OSEBX. In an enhanced index portfolio, we find that only the latter approach yields significant excess returns in FF3. This is a *key insight* when exploiting the index effect in an enhanced index portfolio. This must come from the fact that GLM can only constitute an active share of 2.8%–4.1%, compared to XGBC30 at 22.1%, in an enhanced index portfolio. Finally, we were able to create enhanced index portfolios outperforming OSEBX over the last 13 years, and we conclude that we were able to outperform OSEBX to a reasonable extent.

6 Summary and Conclusions

In this thesis, we have investigated the practical application of exploiting the index effect on OSEBX using ML. We did this by using GLM and XGBoost ML models, predicting upcoming changes to OSEBX, by using data variables from Euronext in the OSEBX index methodology. We simulated a portfolio of trading on ML predictions from 2010 to 2023, in both an active trading and enhanced index strategy. Lastly, we analysed the returns from the simulated portfolios.

Our three research questions were to determine (1) "To what degree can machine learning algorithms predict the index composition of OSEBX in the next period?", (2) "To what degree can trading on predicted additions and deletions outperform OSEBX in an active trading portfolio?", and (3) "To what degree can trading on predicted additions and deletions outperform OSEBX in an enhanced index portfolio?".

(1) We found that it is possible to obtain high accuracy when predicting if a company will be on OSEBX or not in the next period. We obtained more than 95% accuracy simply by using a lagged variable of the target variable. However, it proved to be more difficult when predicting the actual additions and deletions on OSEBX. By using conditional PPT in XGBoost, we were able to increase the total number of true predicted changes, with the trade-off being lower total accuracy.

(2) When simulating the active trading portfolios over time, we found that several portfolios were able to outperform OSEBX, both in terms of R_p , SR, σ_p , α_{CAPM} and α_{FF3} . In short, both the high-risk high-return GLM100 and GLM60 models, and the lower-risk XGBC30 model were able to create excess returns compared to OSEBX. The key insight is that our findings suggest that the index effect can be exploited by two different approaches. We also point out that the introduction of the conditional PPT in the XGBoost Custom models seems to have served its purpose, as the XGBC30 model proved to be significantly less volatile (yet yielding formidable returns) than the competing GLM portfolios.

(3) Lastly, we also found that trading on recommendations from ML models was able to outperform OSEBX in enhanced index portfolios, even when operating within an annual TE of 2%. The GLM portfolios gave tiny active shares in the enhanced index portfolio,

because of their volatile nature. On the other hand, the XGBC30 enhanced index portfolio still performed well and was the only portfolio able to generate alpha in FF3 within a 10% confidence level.

Moreover, we would like to draw a more general conclusion based on our findings in this thesis. Assuming we were able to outperform OSEBX, we want to point out the effect of applying machine learning in practice. We have, as mentioned several times throughout this thesis, found that accuracy is not necessarily the ultimate measurement of assessing model performance. We have in our case shown several times that even the least accurate models can create the most diversified and high-yielding portfolios. The conclusions drawn from this work do therefore come down to the following: one should clearly understand the problem objective before making a predictive ML model. This can in turn broaden the horizon to which one can apply machine learning. If we in this thesis had gone with the model showing the highest accuracy, we would not have been able to exploit anything - because that model predicted no additions nor deletions to OSEBX at all.

6.1 Robustness of Findings

In this thesis, we made several assumptions regarding finance and methodology. As with any findings, the performance is dependent on underlying assumptions. Regarding our findings in the first research question, the robustness will depend on several factors, including the quality of the data set and our understanding of the official OSEBX rule book. In practice, we faced several missing values in the data set, where we then decided to delete the row rather than trying to find the exact data point. Deleting rows from the data frame can in turn decrease robustness, which creates for a weakness in our data set and models.

When it comes to the second and third research questions, there are also several factors to consider. The first one is the financial assumptions made throughout this thesis. We have, as far as we could, tried to use assumptions that mirror real-world trading. There is no doubt that changing the financial assumptions will affect the results presented in this thesis, and whether or not the assumptions represent real-world trading is open for debate. Also, the FF3 regressions come with some setbacks. Firstly, we have used risk factors that do not specifically represent the exact market we defined. This can cause

misleading regression results, and the results should be interpreted carefully. Furthermore, we point out that by using only three risk factors, we are not necessarily capturing the entire risk picture. We could have used several more risk factors in our regressions, but due to the lack of available factors in the Norwegian market, we only used the FF3 model with 3 factors. This is therefore another weakness in our findings.

Regarding the portfolio simulations, we are also facing some issues challenging the robustness. This is mainly due to missing data in the VWAP data frame. When running the simulations, we have gathered data from this data frame to simulate the portfolio development. Several times we have experienced issues with companies (especially in 2010 and 2011) lacking VWAP. This is a limitation to the simulations, as we then have been forced to drop those predictions from the simulations.

Furthermore, we optimised the enhanced index strategy for TE_{ann} at 2%. This is in the upper range for accepted TE for an enhanced index fund. In the enhanced index strategy, a lower TE would in turn mean lower excess returns.

6.2 Practical Recommendations

Practically, one can go further in optimising the trading strategy discussed in [section 5](#), which we have not done in this thesis. Possible improvements can be made for (1) time horizon, (2) ε in the conditional PPT, (3) hyper-parameter tuning, and (4) weighting of the different securities in the portfolios. In sum, trading on the index effect using ML may be optimised further. Despite this, we conclude that our portfolio simulation shows high performance, while still maintaining a low degree of manual intervention and a high degree of repeatability.

6.3 Further Research

Furthermore, this thesis indicates that there *may* be somewhat of a market inefficiency, surrounding the index rebalancing events on OSEBX. We argue that the addition or deletion of a stock does not change the immediate underlying value of the company, so if EMH holds, we should not have been able to exploit the index effect. Therefore it would be of great interest to know if the index effect is a current phenomenon on OSEBX, or if

it will continue to exist in the future with more and more investors being aware of it. To fully understand the index effect, one should therefore investigate other indices, which differ from OSEBX in both market size and maturity.

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Appendix A Abbreviations and their explanations

ABBR	Explanation
AD	Announcement date
AI	Artificial Intelligence
API	Application Programming Interface
AUM	Assets Under Management
AUROC	Area Under Receiver Operating Characteristic
BBG	Bloomberg
CAPM	Capital Asset Pricing Model
CD	Cut-off date
EMH	Efficient Market Hypothesis
EOB	Electronic Order Book
FF	Free Float Market Cap
FF3	Fama-French 3 Factor Model
FN	False Negative
FP	False Positive
FTSE	Financial Times Stock Exchange
GICS	Global Industry Classification Standard
GLM	Generalised Linear Model
HML	High Minus Low
ICB	Industry Classification Benchmark
IR	Information Ratio
LLM	Large Language Model
ML	Machine Learning
NBIM	Norges Bank Investment Management
NHH	Norwegian School of Economics
OLS	Ordinary Least Squares
OSE	Oslo Stock Exchange
OSEAX	Oslo Stock Exchange All Shares Index
OSEBX	Oslo Stock Exchange Benchmark Index
PPT	Posterior Probability Threshold
RF	Random Forest
RL	Reinforcement Learning
ROC	Receiver Operating Characteristic
SL	Supervised Learning
SMB	Small Minus Big
SR	Sharpe Ratio
TE	Tracking Error
TN	True Negative
TO	Turnover Electronic Order Book
TP	True Positive
TS	Trading Status
TSCV	Time Series Cross-Validation
UL	Unsupervised Learning
VIP	Variance Importance Plot
VWAP	Volume Weighted Average Price
XGBB	XGBoost Benchmark
XGBC	XGBoost Custom
XGBoost	eXtreme Gradient Boosting

Table A.1: Abbriviation

Appendix B OSEBX Constituents Since 2006

OSEBX constituents at the start of 2006		
ABG Sundal Collier Holding ASA	Ekornes ASA	Otello Corp ASA
Akastor ASA	Eltek AS	PGS ASA
Aker ASA	Equinor ASA	PRA Group Europe AS
Andvord Tybring-Gjedde	Expert AS	PhotoCure ASA
Arribatec Group ASA	Frontline PLC	Profdoc AS
Atea ASA	Funcom Se	Prosafe SE
Axactor ASA	Hafslund ASA	Q-Free ASA
BW Gas AS	Jason Shipping AS	Royal Caribbean Cruises Ltd
BWG Homes ASA	Kongsberg Automotive ASA	STX Europe AS
Carasent ASA	Leroy Seafood Group ASA	SUBSEA 7 Inc
Cermaq Group AS	Magnora ASA	Schibsted ASA
Crew Gold Corp	Microsoft Dev Center Norway AS	Seadrill Ltd/old
DNB Bank ASA	Mowi ASA	Sinvest ASA
DNO ASA	NEL ASA	Stolt-Nielsen Ltd
Dolphin Drilling ASA	Nera ASA	Storebrand ASA
Norsk Hydro ASA	Ocean RIG ASA	Subsea 7 SA
Norwegian Air Shuttle ASA	Odfjell SE	SuperOffice AS
NorGani Hotels ASA	Odfjell SE	TGS ASA
Origio A/S	Orkla ASA	TOMRA Systems ASA
Tandberg AS	Tandberg Television ASA	Techstep ASA
Telenor ASA	Veidekke ASA	Wilh Wilhelmsen Holding ASA
Wilh Wilhelmsen Holding ASA	Yara International ASA	

Table B.1: OSEBX constituents at the start of 2006

This table shows all companies composing OSEBX in 2006, which is what our data covers.

Appendix C OSEBX changes since 2006

Company	Additions	Deletions
ABG Sundal Collier Holding ASA	2020-05-29	2016-11-30
AF Gruppen ASA	2013-05-31	
AMSC ASA	2019-05-31, 2016-11-30, 2014-05-30	2021-03-19, 2017-05-31, 2016-05-31
Akastor ASA	2015-05-29	
Aker ASA	2011-11-30, 2009-11-30	2010-05-31, 2009-05-29
Aker BP ASA	2012-05-31	
Aker BioMarine ASA	2011-05-31, 2007-06-29	2011-11-30, 2007-12-30
Aker Carbon Capture ASA	2023-03-17, 2022-03-18	2022-09-16
Aker Horizons ASA	2021-09-17	
Aker Solutions ASA	2016-11-30	2016-05-31
Algeta ASA	2009-11-30	
Altinex AS	2006-12-29	
Archer Ltd	2011-05-31	2011-11-30
ArcticZymes Technologies ASA	2021-03-19, 2014-05-30	2017-05-31
Arendals Fossekompagni ASA	2022-03-18	
Arribatec Group ASA	2011-11-30	
Asetek A/S	2016-11-30	2021-03-19, 2014-11-28
Austevoll Seafood ASA	2018-05-31, 2007-06-29	2020-05-29, 2013-11-29
AutoStore Holdings Ltd	2022-03-18	
Avance Gas Holding Ltd	2020-05-29, 2015-05-29	2022-03-18, 2016-11-30
Axactor ASA	2016-05-31	2021-09-17, 2007-06-29
Axel Springer Norway AS	2006-12-29	
BW Gas AS	2007-06-29	
BW LPG Ltd	2017-11-30, 2014-05-30	2016-11-30
BW Offshore Ltd	2018-05-31, 2010-11-30, 2007-06-29	2021-03-19, 2011-05-31, 2008-06-30
BWG Homes ASA	2009-05-29	2008-12-30
Bakkafrost P/F	2014-05-30, 2011-11-30	2013-05-31, 2010-11-30
Bank Norwegian ASA	2017-11-30, 2016-11-30	2017-05-31
Bergenbio ASA	2018-05-31	2023-03-17
Bonheur ASA	2019-11-29	
Borr Drilling Ltd	2023-03-17, 2017-11-30	2020-05-29
Borregaard ASA	2021-03-19	
Bouvet ASA	2020-05-29	
COSL Holding AS	2006-12-29	
Cadeler A/S	2022-09-16	
Carasent ASA	2021-03-19	2023-03-17, 2007-12-30
Cermaq Group AS	2014-05-30	
Circio Holding ASA	2017-11-30	2018-11-30
Cloudberry Clean Energy ASA	2022-03-18	
Copeinca ASA	2009-11-30, 2007-12-30	2010-05-31, 2008-12-30
Crayon Group Holding ASA	2020-05-29	
Crew Gold Corp	2007-06-29	
Data Respons ASA	2019-11-29, 2006-12-29	2009-11-30
Dolphin Drilling ASA	2015-11-30	
Electromagnetic Geoservices ASA	2012-11-30	2013-11-29
Elkem ASA	2019-05-31	
Elmera Group ASA	2019-05-31	
Eltek AS	2010-05-31	2008-12-30
Ensurge Micropower ASA	2015-05-29	2019-05-31
Europris ASA	2015-11-30	
Evry AS	2017-11-30	
FLEX LNG Ltd	2021-09-17	
Fjord1 AS	2018-05-31	2021-03-19
Frontline PLC	2015-05-29	2013-05-31
Funcom Se	2017-11-30, 2012-05-31	2018-11-30, 2012-11-30, 2008-12-30
Gaming Innovation Group Inc	2021-09-17, 2017-05-31, 2016-05-31	2022-09-16, 2021-03-19, 2016-11-30
Gjensidige Forsikring ASA	2011-05-31	
Golden Ocean Group Ltd/Old	2006-12-29	
Grieg Seafood ASA	2017-05-31	2021-09-17
Hafnia Ltd	2022-09-16	
Hafslund ASA	2016-11-30, 2006-12-29	2013-11-29, 2009-05-29
Hexagon Composites ASA	2019-05-31, 2016-05-31, 2014-05-30	2018-11-30, 2014-11-28
IDEX Biometrics ASA	2015-05-29	2021-09-17
IMAREX ASA	2007-12-30	2009-11-30
Itera ASA	2007-06-29	2008-06-30

Continuation from previous page		
Company	Additions	Deletions
Jason Shipping AS	2007-12-30	2009-05-29, 2007-06-29
Jinhui Shipping & Transportation Ltd	2010-11-30, 2007-06-29	2011-05-31, 2008-12-30
Kahoot! ASA	2021-09-17	
Karo Pharma Norge AS	2014-11-28, 2010-05-31	2013-05-31
Kid ASA	2021-03-19	
Kitron ASA	2016-11-30	
Komplett ASA	2006-12-29	2008-12-30
Kongsberg Gruppen ASA	2010-05-31, 2008-06-30	2009-11-30
Leroy Seafood Group ASA	2016-11-30, 2009-11-30	2014-11-28, 2009-05-29
Link Mobility Group ASA	2017-05-31	
MPC Container Ships ASA	2018-11-30	
Magnora ASA	2012-05-31	
Mamut AS	2007-06-29	2010-05-31
Medistim ASA	2020-05-29	2021-09-17
Morpol ASA	2010-11-30	2012-05-31
Multiconsult ASA	2021-09-17, 2015-11-30	2023-03-17, 2017-05-31
NEL ASA	2018-05-31	2007-12-30
NRC Group ASA	2007-06-29	2010-05-31
Next Biometrics Group AS	2016-05-31	2019-11-29
Nordic Semiconductor ASA	2010-05-31, 2007-12-30	2008-12-30
Norwegian Property ASA	2018-05-31	
Nykode Therapeutics ASA	2022-09-16	
Oddfell SE	2010-05-31, 2008-06-30	2014-05-30, 2009-05-29, 2008-12-30, 2007-12-30
Olav Thon Eiendomsselskap ASA	2013-05-31, 2008-06-30, 2006-12-29	2021-03-19, 2008-12-30, 2007-06-29
Origio A/S	2007-12-30	2009-11-30, 2007-06-29
Otello Corp ASA	2018-11-30	
PA Resources AB	2006-12-29	2008-12-30
PCI Biotech Holding ASA	2018-05-31	2022-03-18
PGS ASA	2023-03-17	2021-03-19
PRA Group Europe AS	2011-11-30	2012-05-31, 2010-11-30
Pexip Holding ASA	2021-03-19	2022-09-16
PhotoCure ASA	2015-05-29, 2014-05-30, 2010-05-31	2014-11-28, 2012-11-30, 2008-12-30
Polarcus Ltd	2013-05-31	2014-05-30
Prosafe SE	2016-05-31	
Q-Free ASA	2015-05-29, 2010-05-31	2016-11-30, 2014-11-28, 2006-12-29
Questerre Energy Corp	2017-11-30, 2010-05-31	2018-11-30, 2011-11-30, 2009-11-30
REC Silicon ASA	2021-03-19	2019-11-29
RenoNorden ASA	2015-05-29	2015-11-30
Rieber	Son AS	2006-12-29
2007-06-29		
SAS AB	2016-05-31	2016-11-30
SATS ASA	2020-05-29	2023-03-17
Salmar ASA	2013-11-29, 2007-12-30	2013-05-31
Schibsted ASA	2015-11-30	
Seadrill Ltd/old	2018-05-31	
Sinvest ASA	2006-12-29	
Solon Eiendom ASA	2010-05-31	2015-05-29
Songa Offshore SE	2012-11-30, 2009-11-30, 2007-12-30	2013-05-31, 2011-05-31, 2008-12-30
SpareBank 1 SR-Bank ASA	2017-05-31	
Steen & Stroem AS	2007-06-29	
StrongPoint ASA	2009-05-29	2009-11-30
Team Tankers Management Holding AS	2010-05-31, 2007-06-29	2010-11-30, 2009-05-29
Techstep ASA	2008-12-30	
Teekay Petrojarl AS	2006-12-29	
Thor Medical ASA	2015-05-29	2022-09-16
TietoEVRY Oyj	2020-05-29	2021-03-19
Treasure ASA	2016-11-30	2018-05-31
Tribona ASA	2007-12-30	2008-12-30
Ultimovacs ASA	2021-09-17	
Var Energi ASA	2022-09-16	
Veidekke ASA	2012-05-31, 2009-11-30, 2008-06-30	2010-05-31, 2008-12-30, 2007-12-30
Visolit AS	2006-12-29	2007-06-29
Vizrt Ltd	2006-12-29	2011-05-31
Vow ASA	2021-03-19	2022-03-18
Wallenius Wilhelmsen ASA	2010-11-30	
Wavefield Inseis AS	2007-06-29	2008-12-30
Wilh Wilhelmsen Holding ASA	2011-11-30	2020-05-29, 2018-11-30, 2008-12-30
XXL ASA		2022-03-18

Table C.1: All OSEBX Additions and Deletions since 2006

All changes to OSEBX in the period we are analysing in this thesis.

Appendix D CD, AD, and ED from 2006 to 2023

	CD	AD	ED		CD	AD	ED
1	2023-02-17	2023-03-08	2023-03-17	18	2014-05-02	2014-05-15	2014-05-30
2	2022-08-19	2022-09-07	2022-09-16	19	2013-11-01	2013-11-14	2013-11-29
3	2022-02-18	2022-03-09	2022-03-18	20	2013-05-03	2013-05-14	2013-05-31
4	2021-08-20	2021-09-08	2021-09-17	21	2012-11-02	2012-11-16	2012-11-30
5	2021-02-19	2021-03-10	2021-03-19	22	2012-05-03	2012-05-16	2012-05-31
6	2020-04-30	2020-05-12	2020-05-29	23	2011-11-02	2011-11-11	2011-11-30
7	2019-11-01	2019-11-08	2019-11-29	24	2011-05-03	2011-05-12	2011-05-31
8	2019-05-03	2019-05-09	2019-05-31	25	2010-11-02	2010-11-15	2010-11-30
9	2018-11-02	2018-11-09	2018-11-30	26	2010-05-03	2010-05-12	2010-05-31
10	2018-05-03	2018-05-09	2018-05-31	27	2009-11-02	2009-11-13	2009-11-30
11	2017-11-02	2017-11-10	2017-11-30	28	2009-04-30	2009-05-14	2009-05-29
12	2017-05-03	2017-05-05	2017-05-31	29	2008-12-02	2008-12-11	2008-12-30
13	2016-11-02	2016-11-10	2016-11-30	30	2008-06-02	2008-06-05	2008-06-30
14	2016-05-03	2016-05-12	2016-05-31	31	2007-11-30	2007-12-04	2007-12-30
15	2015-11-02	2015-11-12	2015-11-30	32	2007-06-01	2007-06-08	2007-06-29
16	2015-04-30	2015-05-13	2015-05-29	33	2006-12-01	2006-12-04	2006-12-29
17	2014-10-31	2014-11-13	2014-11-28	34	2006-06-02	2006-06-08	2006-06-30

Table D.1: CD, AD, and ED from 2006 to 2023

A summary of all important dates since 2006.

Appendix E ICB Estimations

	ticker	company	ICB (estimated)
1	1643322D NO Equity	Seadrill Ltd/old	Energy
2	EDRILL NO Equity	Seadrill X ASA	Energy
3	ALX NO Equity	Altinex AS	Energy
4	SIN NO Equity	Sinvest ASA	Energy
5	8185832Q NO Equity	Aker Drilling ASA/Old	Energy
6	APL NO Equity	APL ASA	Energy
7	FRID NO Equity	Saipem Discoverer Invest SARL	Energy
8	APLC NO Equity	APL Advanced Production & Loading PLC	Energy
9	NOV NO Equity	Norsk Vekst AS	Financial Services
10	FSL NO Equity	Fesil AS	Industrial Goods and Services
11	GAS NO Equity	BW Gas AS	Industrial Goods and Services
12	POLI NO Equity	Berry Packaging Norway AS	Industrial Goods and Services
13	DESS NO Equity	Deep Sea Supply ASA	Industrial Goods and Services
14	BHOC NO Equity	B+H Ocean Carriers Ltd	Industrial Goods and Services
15	NEMI NO Equity	Nemi Forsikring AS	Insurance
16	PFI NO Equity	P4 Radio Hele Norge AS	Media
17	SST NO Equity	Steen & Stroem AS	Realestate
18	ATG NO Equity	Andvord Tybring-Gjedde	Retail
19	EXPERT NO Equity	Expert AS	Retail
20	CNS NO Equity	Conseptor ASA	Technology
21	CSG NO Equity	Component Software Group	Technology
22	CAPTU NO Equity	Captura AS	Technology
23	ACTIVE NO Equity	Active 24 ASA	Technology
24	TAT NO Equity	Tandberg Television ASA	Technology
25	NSTAT NO Equity	Norstat ASA	Telecommunications
26	NER NO Equity	Nera ASA	Telecommunications
27	RIC NO Equity	Rica Hotels ASA	Travel and Leisure

Table E.1: Estimated ICB-sectors

Estimated ICB supersectors are estimated for the companies we were unable to obtain this information directly from BBG.

Appendix F XGBoost Hyperparameters

	parameters	value
1	nrounds	50
2	max_depth	1000
3	objective	binary:logistic
4	remaining	default

Table F.1: XGBoost Hyperparameters

XGBoost shows great flexibility compared to other models in the ability to tune hyperparameters when optimising for specific use cases. Note that these parameters are not optimised to maximise accuracy.

Appendix G Excel Bloomberg Add In BQL Formulas

Command	Code
trading_status	=@BQL(A2; "(traded_days/" & F2 & ")*100"; "px=px_last(dates=range(" & @BQL.DATE(D2) & ", "& @BQL.DATE(E2) & ")); "traded_days = count(matches(px,px!=NA)).value"; "total_days=count(px().date).value")
turnover	=@BQL(A2;" tot";"high=sum(first(group(sort(turnover(dates=range(" & @BQL.DATE(C2) & ", " & @BQL.DATE(E2) & ")),order=desc)),12)), sum=sum(Group(turnover(dates=range(" & @BQL.DATE(C2) & ", " & @BQL.DATE(E2) & "))))), tot = sum - high")
icb_supersector	=BDP(A2;"icb_supersector_name")
free_float	=BDH(A2;"cur_mkt_cap";E2)* BDH(A2; "eqy_free_float_pct";E2)

Table G.1: Excel BBG Add In BQL Functions

A2: ticker, C2: one year before E2, D2: start date trading period, E2: end date trading period, and F2: total trading days.

Appendix H Descriptive statistics

Boxplot of log free float

Sorted by values where index_cp is 1

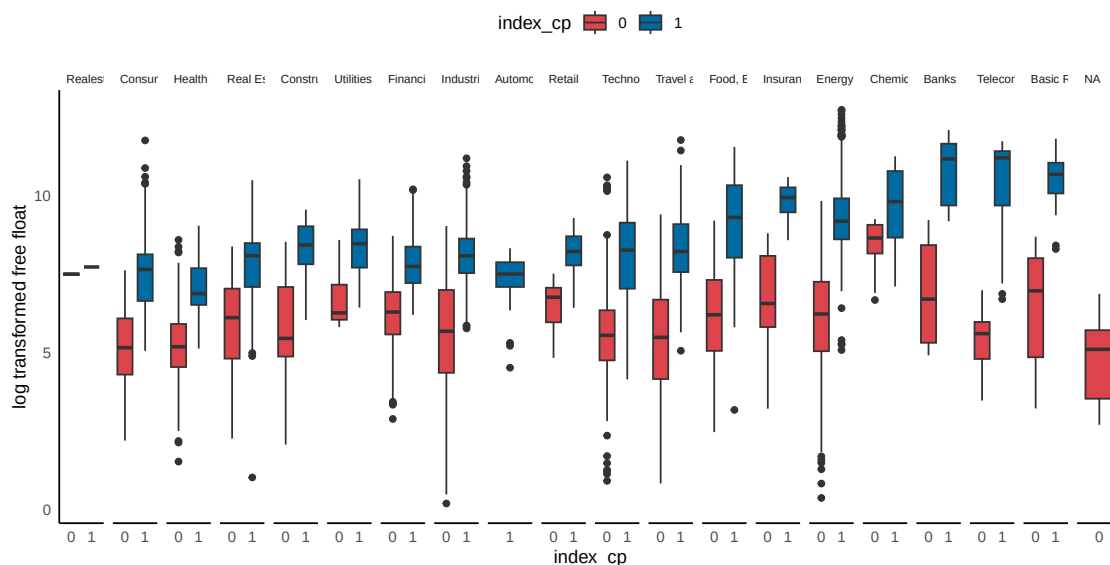


Figure H.1: Box-plot free_float and index_cp

Illustration of allocation of index_cp per different ICB supersector, in relation to the logarithm of free_float.

Boxplot of log turnover

Sorted by values where index_cp is 1

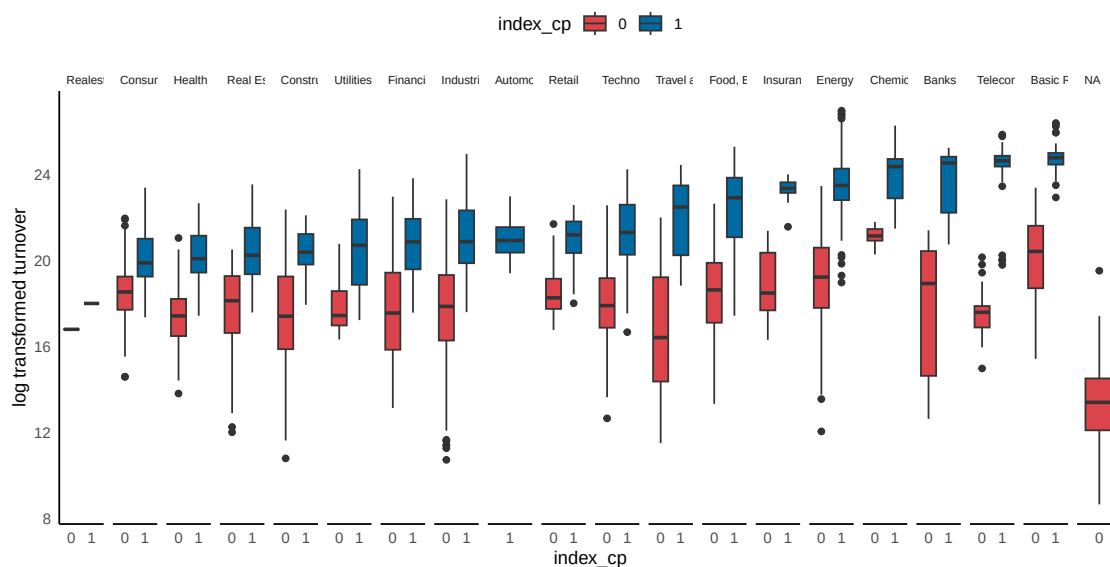


Figure H.2: Box-plot turnover_eob and index_cp

Illustration of allocation of index_cp per different ICB supersector, in relation to the logarithm of turnover_eob.

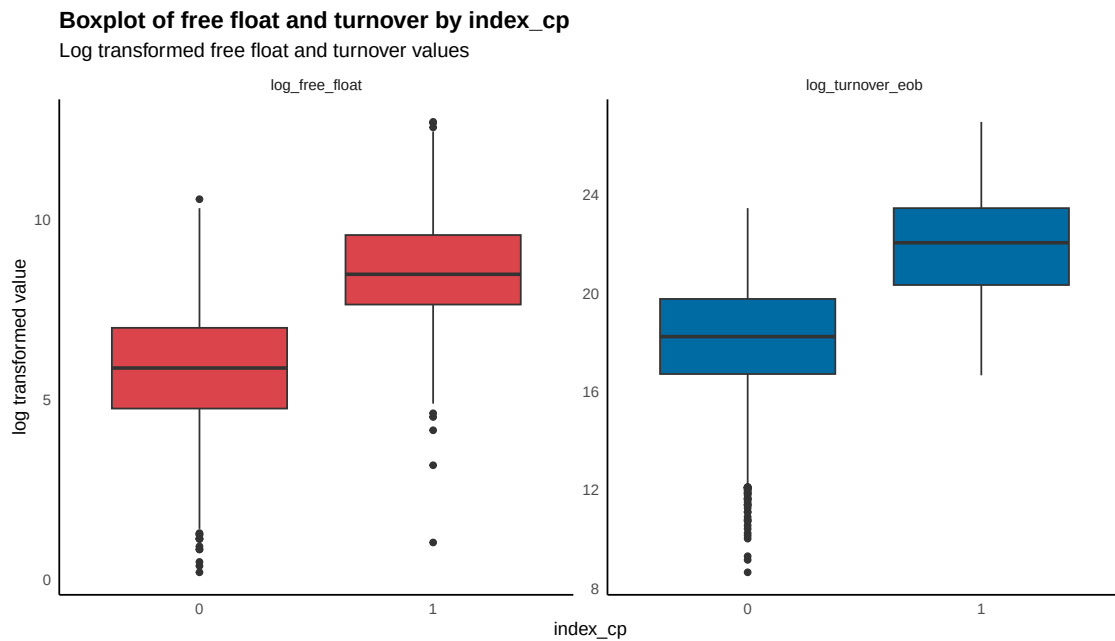


Figure H.3: Box-plot turnover_eob, free_float, and index_cp

Boxplot of free_float and index_cp, and turnover_eob and index_cp

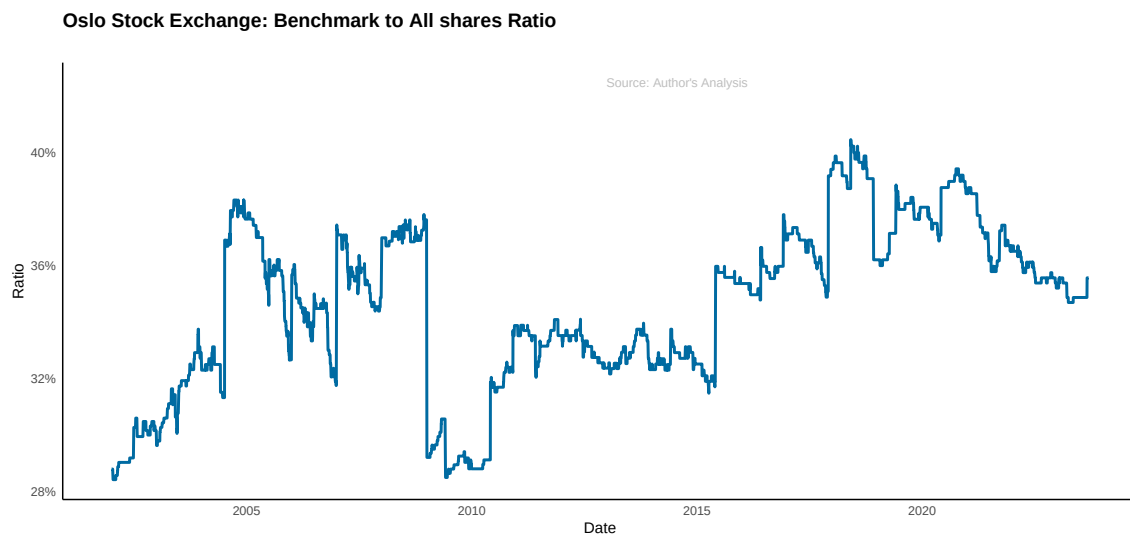


Figure H.4: Ratio of OSEBX to OSEAX member count

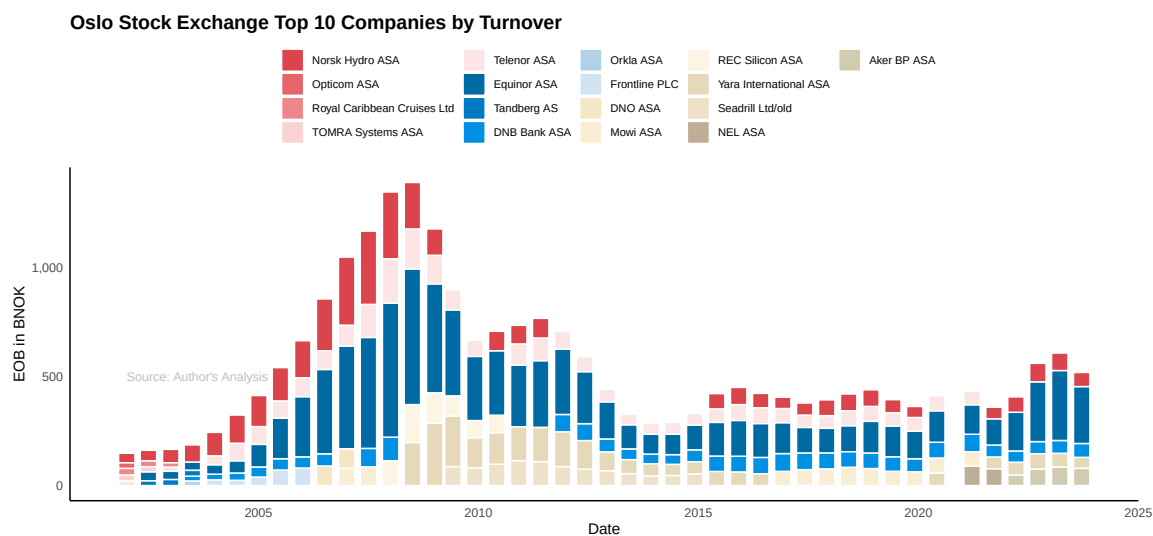


Figure H.5: Turnover of 10 largest OSEBX companies over time

Turnover of 10 largest OSEBX companies over time, being large Norwegian Companies.

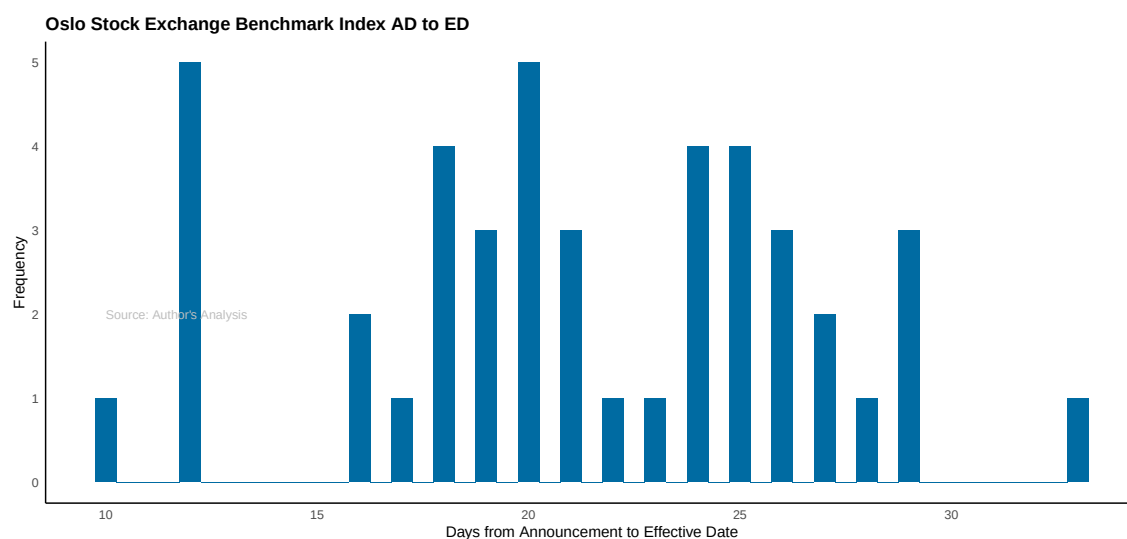


Figure H.6: OSEBX number of days from AD to ED

Number of days between each announcement date and effective date illustrated as a histogram. The length varies between 10 and 32 days.

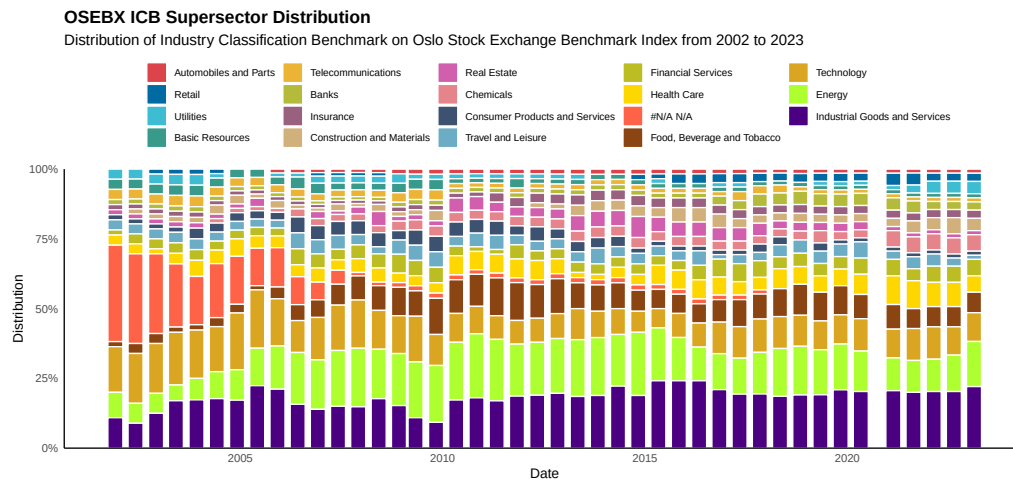


Figure H.7: OSEBX ICB Supersector Distribution over Time

Industry classification benchmark (ICB) for companies on OSEBX in the period from 2002 to 2023. Notice that industry and energy companies dominate OSEBX.

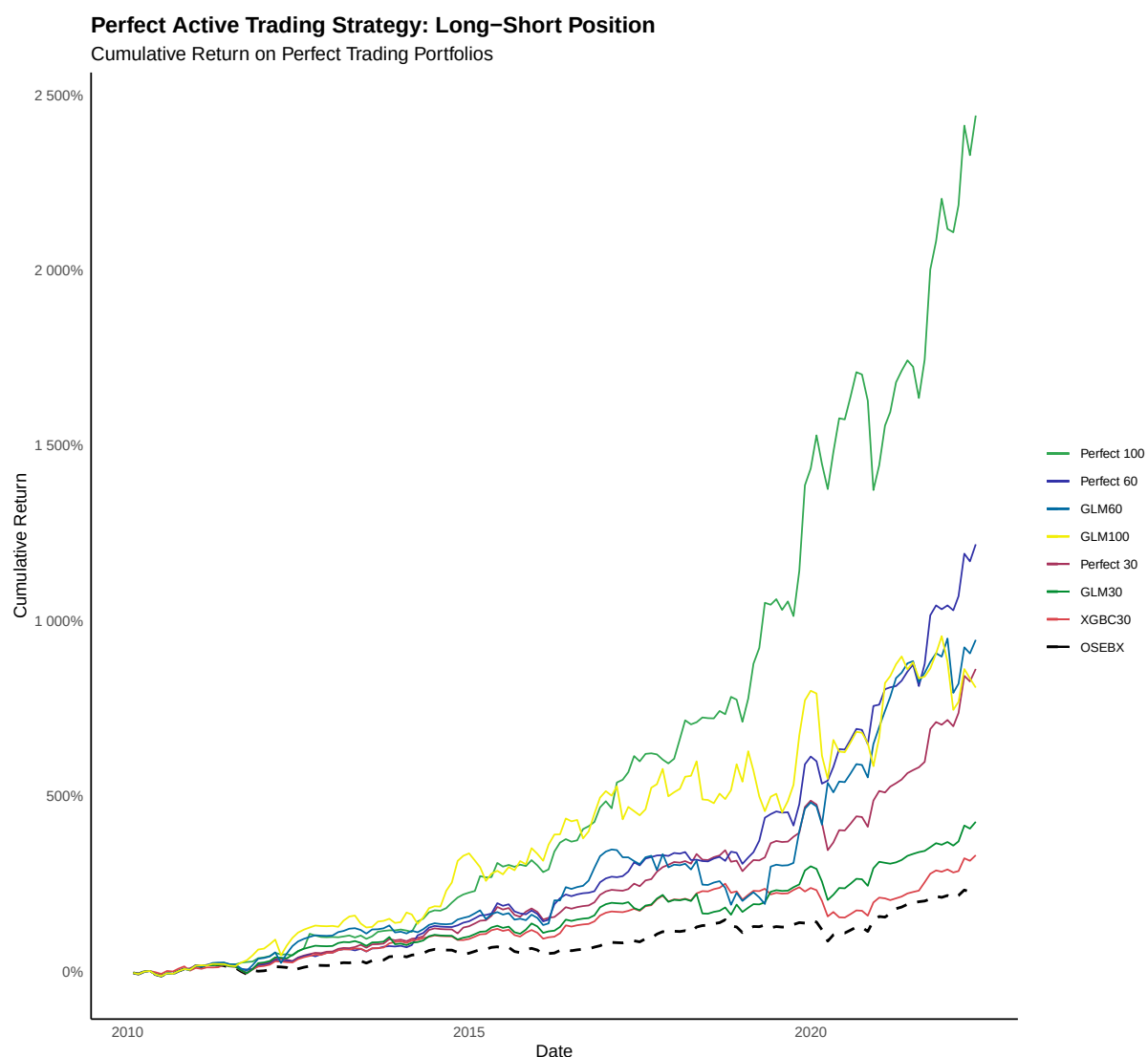


Figure H.8: Cumulative Return Perfect Portfolios

In this figure we have included the portfolios Perfect 30, Perfect 60, and Perfect 100. Those portfolios represent the development if one had known 100% of which companies should enter/exit OSEBX since 2010. Investing NOK 1000 on January 4th, 2010 would have grown to 25,000 on May 31st, 2022, if one knew the changes 100 days in advance of the ED, beating OSEBX by thousands of percent in cumulative return.

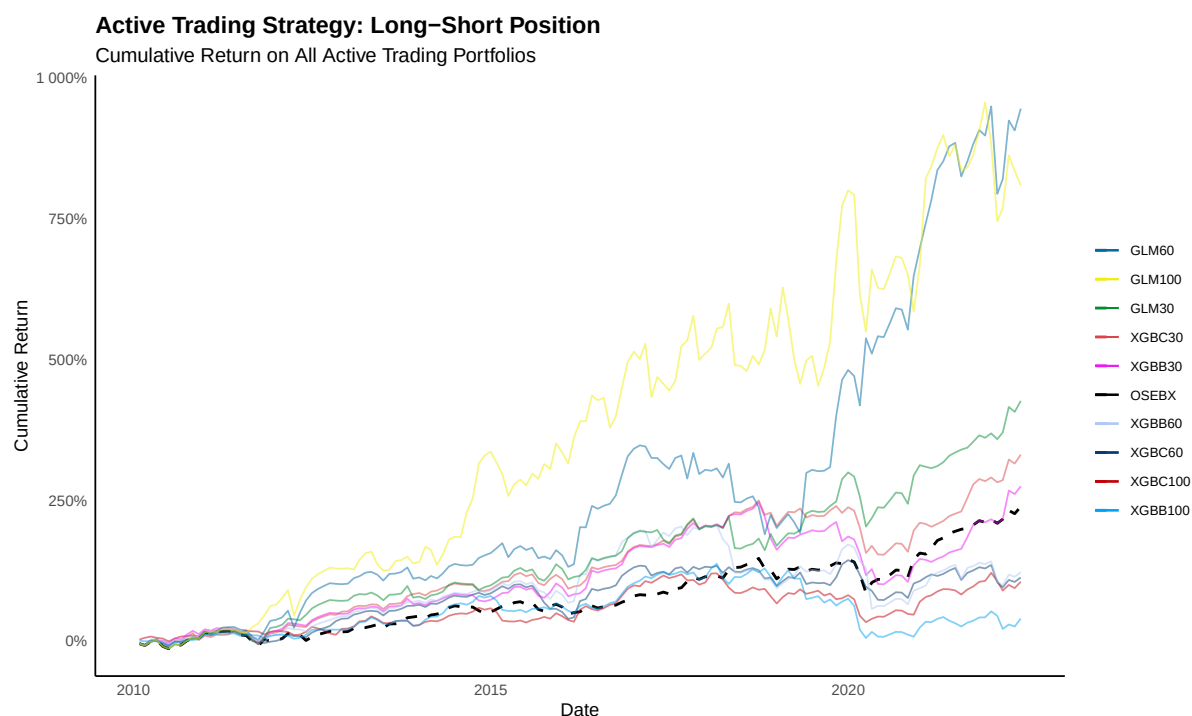


Figure H.9: Cumulative Return Long–Short Trading Strategy

All long-short portfolios simulated are illustrated.



Figure H.10: Cumulative Return Long–Only Trading Strategy

All long-only portfolios simulated are illustrated.

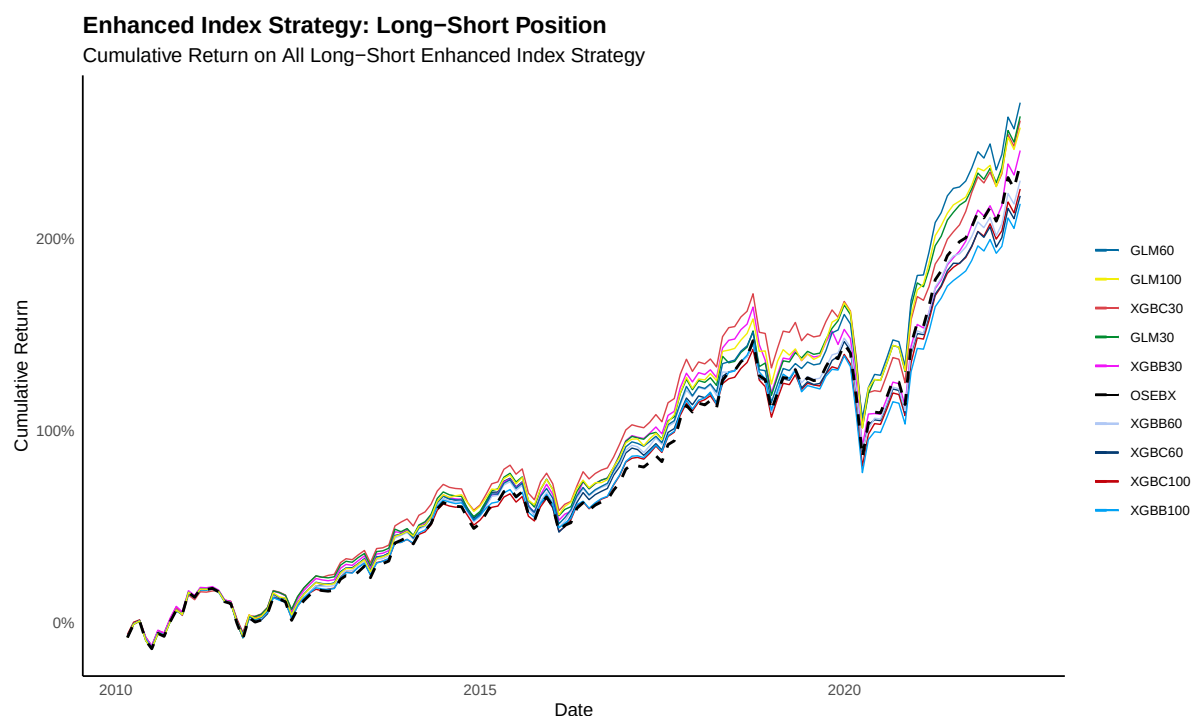


Figure H.11: Cumulative Return Long-Short Enhanced Index

All long-short enhanced portfolios simulated are illustrated.

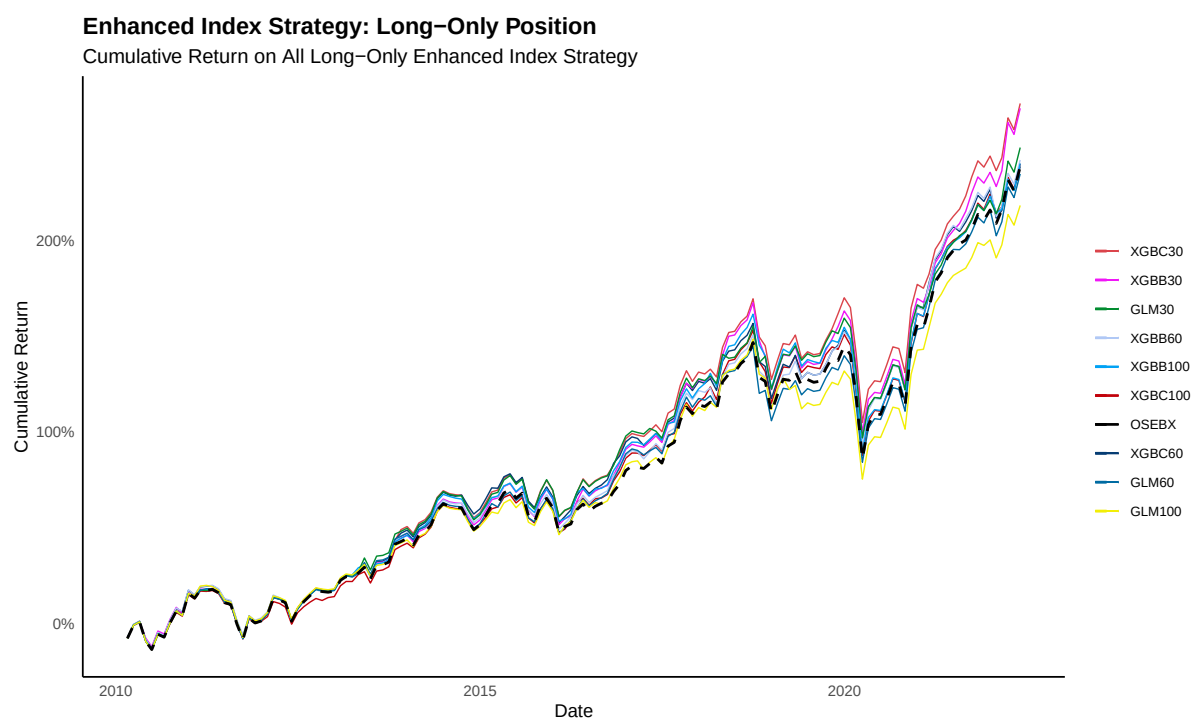


Figure H.12: Cumulative Return Long-Only Enhanced Index

All long-only enhanced portfolios simulated are illustrated.

Appendix I Portfolio Simulation Algorithm

The algorithm used when performing the portfolio simulations can be summarised as the following. All predictions come from the ML models, which are stored in a data frame including the dates of the start and end of the trading periods, and which companies are held in each trading period.

Step 1: Add OSEBX as a long position between each active trading period.

Step 2: Find VWAP for each company for each trading period start and end date, including all periods OSEBX is traded.

Step 3: Remove any rows with NA-values in VWAP (most present for companies in 2010 and 2011).

Step 4: Calculate the return on each trade, illustrated in [Equation I.1](#) where r_l is the return on a long position, r_s is the return on short positions, p_0 is the VWAP at the start of the trading period and p_1 is the VWAP at the end of the trading period. S_{rate} is the annual short rate, and d is the number of days per active trading period (30, 60, or 100 days).

$$r_l = \frac{p_1 - p_0}{p_0}, \quad r_s = \frac{p_0 - p_1}{p_0} - \frac{S_{rate}^{ann} * d}{365} \quad (\text{I.1})$$

Step 5: Calculate the average return for each trading period, as we assume weights between the assets to be equally weighted.

Step 6: Define an amount of money to start with (we used 10 million). For each trading period, multiply the money at the end of the last period by the average return in that period plus 1, minus 8 bps (as we use 4 bps as transaction cost per trade, but at the end of each period we both sell the previous holdings and buy new holdings, therefore 8 bps).

Appendix J Example of Portfolio Simulation

We decided to include the GLM100 portfolio as an example of a simulated portfolio. Please note that this simulation only yields portfolio size at the end of the trading periods, so in practice, we have calculated the return for each day and extracted the return from the same dates as the FF3 factors are reported (being the last date in each month). The point of this Appendix is to illustrate where the exceptional (yet risky) returns come from.

Start	End	Company	$Y_{i,t}$	$\hat{Y}_{i,t}$	$Y_{i,t-1}$	VWAP start	VWAP end	Return
2010-01-04	2011-01-08	OSEBX				380.15	439.92	0.16
2011-01-11	2011-05-31	AKER NO Equity	0	1	0	141.72	153.58	0.08
2011-01-11	2011-05-31	ELT NO Equity	1	0	1	1.00	1.00	-0.01
2011-01-11	2011-05-31	JIN NO Equity	0	0	1	19.86	16.40	0.16
2011-01-11	2011-05-31	QFR NO Equity	1	0	1	16.90	17.87	-0.07
2011-06-01	2011-07-10	OSEBX				437.39	421.46	-0.04
2011-07-13	2011-11-30	MGN NO Equity	1	0	1	11.21	5.14	0.53
2011-07-13	2011-11-30	QEC NO Equity	0	0	1	5.56	3.97	0.28
2011-12-01	2012-01-09	OSEBX				378.59	389.09	0.03
2012-01-12	2012-05-31	MGN NO Equity	0	0	1	6.41	5.21	0.18
2012-06-01	2013-01-08	OSEBX				377.66	456.04	0.21
2013-01-11	2013-05-31	EMGS NO Equity	1	0	1	214.05	208.01	0.02
2013-01-11	2013-05-31	SONG NO Equity	0	0	1	1.00	1.00	-0.01
2013-06-01	2013-07-09	OSEBX				491.71	482.30	-0.02
2013-07-12	2013-11-29	BRG NO Equity	0	1	0	26.89	26.65	-0.01
2013-07-12	2013-11-29	AFG NO Equity	1	0	1	61.82	68.07	-0.11
2013-07-12	2013-11-29	EMGS NO Equity	0	0	1	198.11	157.16	0.20
2013-07-12	2013-11-29	PLCS NO Equity	1	0	1	432.09	409.77	0.04
2013-11-30	2014-01-07	OSEBX				542.79	547.04	0.01
2014-01-10	2014-05-30	PLCS NO Equity	0	0	1	417.46	341.25	0.17
2014-05-31	2014-07-08	OSEBX				605.26	619.53	0.02
2014-07-11	2014-11-28	ASTK NO Equity	0	0	1	18.94	9.14	0.51
2014-11-29	2015-01-06	OSEBX				566.34	569.98	0.01
2015-01-09	2015-05-29	LSG NO Equity	0	1	0	28.18	25.26	-0.10
2015-05-30	2015-07-10	OSEBX				645.68	634.87	-0.02
2015-07-13	2015-11-30	LSG NO Equity	0	1	0	27.35	31.87	0.17
2015-07-13	2015-11-30	SCHB NO Equity	1	1	0	184.81	223.98	0.21
2015-12-01	2016-01-09	OSEBX				632.46	563.75	-0.11
2016-01-12	2016-05-31	LSG NO Equity	0	1	0	31.95	42.64	0.33
2016-06-01	2016-07-10	OSEBX				611.63	610.32	-0.00
2016-07-13	2016-11-30	AUSS NO Equity	0	1	0	73.86	80.46	0.09
2016-07-13	2016-11-30	LSG NO Equity	1	1	0	41.26	47.08	0.14
2016-12-01	2017-01-08	OSEBX				662.79	694.57	0.05
2017-01-11	2017-05-31	AUSS NO Equity	0	1	0	77.41	71.05	-0.08
2017-01-11	2017-05-31	GSF NO Equity	1	1	0	72.68	63.21	-0.13
2017-06-01	2017-07-10	OSEBX				713.19	702.25	-0.02
2017-07-13	2017-11-30	BANO NO Equity	1	1	0	81.12	95.05	0.17
2017-07-13	2017-11-30	AUSS NO Equity	0	1	0	68.92	70.13	0.02
2017-12-01	2018-01-08	OSEBX				798.41	834.73	0.05
2018-01-11	2018-05-31	QEC NO Equity	1	0	1	6.11	5.70	0.06

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Start	End	Company	$Y_{i,t}$	$\hat{Y}_{i,t}$	$Y_{i,t-1}$	VWAP start	VWAP end	Return
2018-01-11	2018-05-31	1643322D NO Equity	0	0	1	581.85	952.72	-0.65
2018-01-11	2018-05-31	B2I NO Equity	0	1	0	20.69	18.25	-0.12
2018-01-11	2018-05-31	AUSS NO Equity	1	1	0	65.94	97.46	0.48
2018-06-01	2018-07-10	OSEBX				880.53	899.13	0.02
2018-07-13	2018-11-30	QEC NO Equity	0	0	1	3.05	2.56	0.15
2018-12-01	2019-01-08	OSEBX				860.98	836.40	-0.03
2019-01-11	2019-05-31	RECSI NO Equity	1	0	1	6.57	4.93	0.24
2019-01-11	2019-05-31	SASNO NO Equity	0	1	0	5.86	3.22	-0.45
2019-06-01	2019-07-09	OSEBX				852.09	883.18	0.04
2019-07-12	2019-11-29	RECSI NO Equity	0	0	1	4.67	2.68	0.42
2019-11-30	2020-01-07	OSEBX				902.45	937.40	0.04
2020-01-10	2020-05-29	ASA NO Equity	0	1	0	122.82	111.89	-0.09
2020-01-10	2020-05-29	HAFNI NO Equity	0	1	0	28.65	17.15	-0.40
2020-01-10	2020-05-29	TIETO NO Equity	1	1	0	281.33	252.93	-0.10
2020-05-30	2020-10-27	OSEBX				796.77	826.00	0.04
2020-10-30	2021-03-19	BRG NO Equity	1	1	0	127.72	184.24	0.44
2020-10-30	2021-03-19	AUSS NO Equity	0	1	0	64.58	102.50	0.59
2020-10-30	2021-03-19	KAHOT NO Equity	0	1	0	1.00	1.00	0.00
2020-10-30	2021-03-19	PEXIP NO Equity	1	1	0	1.00	1.00	0.00
2021-03-20	2021-04-27	OSEBX				1050.54	1084.14	0.03
2021-04-30	2021-09-17	ASA NO Equity	0	1	0	84.45	36.71	-0.57
2021-04-30	2021-09-17	AUSS NO Equity	0	1	0	106.73	105.64	-0.01
2021-04-30	2021-09-17	ACC NO Equity	0	1	0	16.93	25.20	0.49
2021-04-30	2021-09-17	KAHOT NO Equity	1	1	0	87.07	70.11	-0.19
2021-04-30	2021-09-17	TIETO NO Equity	0	1	0	288.06	282.25	-0.02
2021-04-30	2021-09-17	SASNO NO Equity	0	1	0	1.97	1.95	-0.01
2021-09-18	2021-10-26	OSEBX				1141.17	1215.74	0.07
2021-10-29	2022-03-18	AUSS NO Equity	0	1	0	114.14	131.70	0.15
2021-10-29	2022-03-18	ACC NO Equity	1	1	0	31.04	19.44	-0.37
2021-10-29	2022-03-18	AUTO NO Equity	1	1	0	33.69	34.39	0.02
2021-10-29	2022-03-18	TIETO NO Equity	0	1	0	260.48	241.40	-0.07
2022-03-19	2022-04-26	OSEBX				1228.54	1226.68	-0.00
2022-04-29	2022-09-16	VAR NO Equity	1	1	0	40.24	37.86	-0.06
2022-04-29	2022-09-16	AUSS NO Equity	0	1	0	149.91	100.30	-0.33
2022-04-29	2022-09-16	GSF NO Equity	0	1	0	140.78	108.44	-0.23
2022-04-29	2022-09-16	NYKD NO Equity	1	1	0	37.33	34.20	-0.08
2022-04-29	2022-09-16	TIETO NO Equity	0	1	0	236.26	265.75	0.12
2022-09-17	2022-10-25	OSEBX				1175.15	1139.47	-0.03
2022-10-28	2023-03-17	BORR NO Equity	1	1	0	46.43	71.26	0.53
2022-10-28	2023-03-17	PGS NO Equity	1	1	0	6.68	9.85	0.47
2022-10-28	2023-03-17	TIETO NO Equity	0	1	0	245.46	318.92	0.30

Table J.1: Explanation of the returns for portfolio GLM100

In the table, we show all simulated trades from the period January 4th 2010 to March 17th, 2023. The simulated trades are based on the predictions given by the GLM100 model. The prediction is shown in column $\hat{Y}_{i,t}$, whereas the column $Y_{i,t}$ shows the true classification. Here we also use the column $Y_{i,t-1}$, as we used this variable to filter out only the cases where the prediction was different from the classification in the previous period. We decided to show the development of the GLM100 portfolios because it (1) has the fewest trades and is most suited to be presented, and (2), it has exceptional returns. Following the simulation algorithm presented in [Appendix I](#), this table shows the simulation after Step 4.

Please see the next page for steps 5 and step 6 from the algorithm presented in [Appendix I](#).

	Start	End	Period Growth Factor	Money at start	Money at end	Cumulative Return
1	2010-01-04	2011-01-08	1.16	10000000.00	11564121.90	0.16
2	2011-01-11	2011-05-31	1.04	11564121.90	12040012.19	0.20
3	2011-06-01	2011-07-10	0.96	12040012.19	11591875.89	0.16
4	2011-07-13	2011-11-30	1.40	11591875.89	16252111.64	0.63
5	2011-12-01	2012-01-09	1.03	16252111.64	16689853.95	0.67
6	2012-01-12	2012-05-31	1.18	16689853.95	19619048.30	0.96
7	2012-06-01	2013-01-08	1.21	19619048.30	23675113.39	1.37
8	2013-01-11	2013-05-31	1.00	23675113.39	23732143.26	1.37
9	2013-06-01	2013-07-09	0.98	23732143.26	23258988.48	1.33
10	2013-07-12	2013-11-29	1.03	23258988.48	23912763.95	1.39
11	2013-11-30	2014-01-07	1.01	23912763.95	24080868.67	1.41
12	2014-01-10	2014-05-30	1.17	24080868.67	28195239.50	1.82
13	2014-05-31	2014-07-08	1.02	28195239.50	28837432.46	1.88
14	2014-07-11	2014-11-28	1.51	28837432.46	43421198.07	3.34
15	2014-11-29	2015-01-06	1.01	43421198.07	43665539.34	3.37
16	2015-01-09	2015-05-29	0.90	43665539.34	39106001.69	2.91
17	2015-05-30	2015-07-10	0.98	39106001.69	38420002.67	2.84
18	2015-07-13	2015-11-30	1.19	38420002.67	45635519.06	3.56
19	2015-12-01	2016-01-09	0.89	45635519.06	40641200.64	3.06
20	2016-01-12	2016-05-31	1.33	40641200.64	54206635.57	4.42
21	2016-06-01	2016-07-10	1.00	54206635.57	54047169.52	4.40
22	2016-07-13	2016-11-30	1.12	54047169.52	60230569.96	5.02
23	2016-12-01	2017-01-08	1.05	60230569.96	63070370.40	5.31
24	2017-01-11	2017-05-31	0.89	63070370.40	56320039.45	4.63
25	2017-06-01	2017-07-10	0.98	56320039.45	55411060.44	4.54
26	2017-07-13	2017-11-30	1.09	55411060.44	60610764.94	5.06
27	2017-12-01	2018-01-08	1.05	60610764.94	63319485.01	5.33
28	2018-01-11	2018-05-31	0.94	63319485.01	59596047.59	4.96
29	2018-06-01	2018-07-10	1.02	59596047.59	60807256.29	5.08
30	2018-07-13	2018-11-30	1.15	60807256.29	69864846.01	5.99
31	2018-12-01	2019-01-08	0.97	69864846.01	67814392.22	5.78
32	2019-01-11	2019-05-31	0.89	67814392.22	60578852.13	5.06
33	2019-06-01	2019-07-09	1.04	60578852.13	62740714.85	5.27
34	2019-07-12	2019-11-29	1.42	62740714.85	88741985.22	7.87
35	2019-11-30	2020-01-07	1.04	88741985.22	92107783.02	8.21
36	2020-01-10	2020-05-29	0.80	92107783.02	73878509.25	6.39
37	2020-05-30	2020-10-27	1.04	73878509.25	76529685.22	6.65
38	2020-10-30	2021-03-19	1.26	76529685.22	96169292.26	8.62
39	2021-03-20	2021-04-27	1.03	96169292.26	99168192.32	8.92
40	2021-04-30	2021-09-17	0.95	99168192.32	93929746.38	8.39
41	2021-09-18	2021-10-26	1.07	93929746.38	99992462.13	9.00
42	2021-10-29	2022-03-18	0.93	99992462.13	93104559.04	8.31
43	2022-03-19	2022-04-26	1.00	93104559.04	92889115.82	8.29
44	2022-04-29	2022-09-16	0.88	92889115.82	82061518.12	7.21
45	2022-09-17	2022-10-25	0.97	82061518.12	79504310.41	6.95
46	2022-10-28	2023-03-17	1.44	79504310.41	114120732.94	10.41

Table J.2: Trading period returns for portfolio GLM100

In this table, we complete steps 5 and 6, meaning we first have taken the average return per trade plus 1 per period "Period Growth Factor", and we have multiplied the factor by "Money at start" for each period. Here, the transaction fees are subtracted from the growth factor before multiplying with the amount of money. Please notice that the simulated portfolio uses a longer period than the period analysed in the thesis.

Appendix K FF3 Regressions against OSEBX

	<i>Dependent variable:</i>									
	GLML			XGBBL			XGBCL			
	30	60	100	30	60	100	30	60	100	OSEBX
OSEBX	0.980*** (0.011)	0.964*** (0.011)	0.969*** (0.011)	0.983*** (0.011)	0.952*** (0.011)	0.934*** (0.010)	0.977*** (0.011)	0.953*** (0.011)	0.919*** (0.009)	1.000*** (0.000)
SMB	−0.019 (0.012)	−0.005 (0.011)	−0.006 (0.012)	−0.011 (0.012)	−0.006 (0.011)	0.013 (0.011)	−0.022* (0.012)	−0.005 (0.011)	0.012 (0.010)	−0.000 (0.000)
HML	−0.005 (0.009)	−0.014* (0.008)	−0.015* (0.008)	0.010 (0.009)	0.010 (0.008)	0.007 (0.008)	0.003 (0.009)	0.008 (0.008)	0.005 (0.007)	−0.000 (0.000)
Constant	0.001* (0.001)	0.001 (0.0005)	0.001 (0.0005)	0.001 (0.001)	0.0003 (0.0005)	−0.0001 (0.0005)	0.001* (0.001)	0.0001 (0.0005)	0.0001 (0.0004)	0.000*** (0.000)
OBS	148	148	148	148	148	148	148	148	148	148
R ²	0.982	0.982	0.982	0.982	0.982	0.984	0.982	0.982	0.986	1.000
AdjR ²	0.981	0.982	0.982	0.981	0.982	0.983	0.982	0.982	0.985	1.000

Note:

*p<0.1; **p<0.05; ***p<0.01

Table K.1: FF3 Regression: Long-Short Enhanced Portfolios against OSEBX

The table shows the regression summary against OSEBX. This is the enhanced long-short portfolios.

	<i>Dependent variable:</i>									
	GLM			XGBB			XGBC			
	30	60	100	30	60	100	30	60	100	OSEBX
OSEBX	0.995*** (0.011)	1.005*** (0.011)	0.997*** (0.011)	0.993*** (0.011)	0.983*** (0.011)	0.981*** (0.011)	0.991*** (0.011)	0.990*** (0.011)	0.969*** (0.011)	1.000*** (0.000)
SMB	−0.010 (0.012)	0.013 (0.012)	0.017 (0.012)	0.011 (0.012)	0.014 (0.012)	0.025** (0.012)	0.008 (0.012)	0.013 (0.012)	0.023* (0.012)	−0.000 (0.000)
HML	0.005 (0.009)	−0.006 (0.009)	−0.005 (0.009)	0.014 (0.009)	0.003 (0.009)	0.005 (0.009)	0.006 (0.009)	0.0003 (0.009)	0.003 (0.009)	−0.000 (0.000)
Constant	0.0004 (0.001)	−0.0003 (0.001)	−0.001 (0.001)	0.001 (0.001)	0.00003 (0.001)	−0.0002 (0.001)	0.001 (0.001)	−0.0001 (0.001)	−0.0001 (0.001)	0.000*** (0.000)
OBS	148	148	148	148	148	148	148	148	148	148
R ²	0.982	0.982	0.982	0.982	0.982	0.982	0.982	0.982	0.982	1.000
AdjR ²	0.981	0.982	0.982	0.981	0.981	0.981	0.981	0.981	0.981	1.000

Note:

*p<0.1; **p<0.05; ***p<0.01

Table K.2: FF3 Regression: Long-Only Enhanced Portfolios against OSEBX

The table shows the regression summary against OSEBX. This is the enhanced long-only portfolios.

	<i>Dependent variable:</i>									
	GLML			XGBBL			XGBCL			
	30	60	100	30	60	100	30	60	100	OSEBX
OSEBX	0.872*** (0.071)	0.562*** (0.143)	0.469*** (0.146)	0.900*** (0.060)	0.495*** (0.113)	0.419*** (0.101)	0.901*** (0.044)	0.690*** (0.076)	0.307*** (0.084)	1.000*** (0.000)
SMB	−0.130* (0.074)	−0.115 (0.149)	−0.109 (0.152)	−0.063 (0.063)	−0.044 (0.118)	0.160 (0.105)	−0.085* (0.046)	−0.027 (0.079)	0.122 (0.088)	−0.000 (0.000)
HML	−0.042 (0.054)	−0.213* (0.109)	−0.142 (0.112)	0.048 (0.046)	0.083 (0.087)	0.005 (0.077)	0.009 (0.034)	0.037 (0.058)	0.014 (0.064)	−0.000 (0.000)
Constant	0.006* (0.003)	0.013** (0.006)	0.014** (0.007)	0.003 (0.003)	0.004 (0.005)	−0.003 (0.005)	0.004* (0.002)	0.0005 (0.003)	0.0001 (0.004)	0.000*** (0.000)
Obs.	147	147	147	147	147	147	147	147	147	147
R ²	0.515	0.109	0.072	0.621	0.131	0.127	0.748	0.378	0.103	1.000
AdjR ²	0.505	0.090	0.052	0.613	0.113	0.109	0.742	0.365	0.084	1.000

Note:

*p<0.1; **p<0.05; ***p<0.01

Table K.3: FF3 Regression: Long-Short Active Portfolios against OSEBX

The table shows the regression summary against OSEBX. This is the active long-short portfolios.

	<i>Dependent variable:</i>									
	GLM			XGBB			XGBC			
	30	60	100	30	60	100	30	60	100	OSEBX
OSEBX	0.984*** (0.039)	1.029*** (0.068)	0.991*** (0.103)	0.958*** (0.057)	0.882*** (0.078)	0.741*** (0.118)	0.953*** (0.052)	0.946*** (0.061)	0.652*** (0.116)	1.000*** (0.000)
SMB	−0.032 (0.040)	0.068 (0.071)	0.192* (0.107)	0.049 (0.059)	0.096 (0.081)	0.302** (0.123)	0.037 (0.054)	0.074 (0.064)	0.204* (0.121)	−0.000 (0.000)
HML	0.016 (0.030)	−0.041 (0.052)	−0.076 (0.079)	0.059 (0.043)	0.023 (0.060)	0.033 (0.090)	0.021 (0.040)	−0.003 (0.047)	0.041 (0.089)	−0.000 (0.000)
Constant	0.002 (0.002)	−0.001 (0.003)	−0.006 (0.005)	0.004 (0.003)	0.001 (0.004)	−0.0004 (0.005)	0.003 (0.002)	−0.0002 (0.003)	0.002 (0.005)	0.000*** (0.000)
Obs.	147	147	147	147	147	147	147	147	147	147
R ²	0.822	0.620	0.409	0.678	0.487	0.255	0.709	0.635	0.207	1.000
AdjR ²	0.818	0.612	0.397	0.672	0.476	0.239	0.703	0.627	0.190	1.000

Note:

*p<0.1; **p<0.05; ***p<0.01

Table K.4: FF3 Regression: Long-Only Active Portfolios against OSEBX

The table shows the regression summary against OSEBX. This is the active long-only portfolios.